THIS IS A SYNTHETIC WORLD: THREE ESSAYS ON CAUSAL INFERENCE, APPLIED ECONOMETRICS, AND MACHINE LEARNING FOR POLICY ANALYSIS

by

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Under the Direction of Jason Coupet, PhD

ABSTRACT

Whether we mean planning, justifying, and implementing a program for a non-profit or evaluating an already existing policy, causal inference to estimate the efficacy of policy is critical to decision making in government, business, and beyond. In general, policy analysts cannot conduct randomized trials, given their cost, ethical status, or infeasibility, and must therefore resort to carefully tuned econometric models to estimate counterfactuals, or a hypothetical scenario of how an outcome would have looked absent a given treatment or policy. Synthetic control methods (SCM) and difference-in-differences (DID) are among the most popular methods of estimating counterfactuals for policy analysts, and have seen much development in application and theory in recent years. In particular, some of their growth can be attributed to their extension via the integration of machine-learning methods. As such, my dissertation consists of three papers under a single theme: causal policy analysis in the high-dimensional setting, where *high-dimensional* refers to where we have many more control units relative to pre-treatment periods.

INDEX WORDS: Synthetic Control, Difference-in-Differences, Causal Inference, Policy Evaluation, Machine Learning

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DEDICATION

For my family, friends, and mentors.

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CHAPTER 1

Paradise Lost? Separating Policy from Pandemic in Hawaii's Border Closure with Single Proxy Synthetic Control

ABSTRACT

Hawaii's March 2020 mandatory traveler quarantine sealed the state's borders at the same instant the COVID-19 pandemic collapsed global travel demand. For a single treated unit these two forces are perfectly collinear in time, so the conventional synthetic-control toolkit—which matches the treated series to a counterfactual that is itself battered by the pandemic—cannot separate the border-closure *policy* from the coincident *pandemic*. I address this with Single Proxy Synthetic Control (SPSC), a member of the proximal / negative-control family that does not attempt to remove the latent pandemic factor but instead *instruments* it using insulated employment sectors that carry the pandemic's macro-recession shock yet are immune to the border policy. Contrasting SPSC against a naive Synthetic Historical Control baseline shows that a substantial share of the measured tourism collapse is pandemic contamination the naive design cannot net out. SPSC recovers a large, robust tourism-demand collapse (roughly 60–75 percentage points of year-over-year growth for the visitor-volume series) but, because the insulated sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly; the estimates from insulated donors alone are therefore best read as *upper bounds in magnitude* on the policy effect and are reported as a robustness band rather than as points. Recovering an actual point requires spanning the missing travel-demand factor from outside the treated economy: adding external travel-demand donors—inbound air travel

to Florida and other exposed-but-open leisure destinations that were not locked down—pulls the Visitor Days estimate off its upper bound, and a convergent ensemble of such donors brackets it in a policy-centered band, taking the step from a bound to a band. The paper demonstrates how negative-control inference disciplines causal claims when a policy and a global shock arrive together.

1.1 Introduction

Among all U.S. states, Hawaii implemented perhaps the most extreme and comprehensive economic shutdown in response to the COVID-19 pandemic. This paper studies the short-run economic consequences of Hawaii's March 2020 mandatory quarantine (Yan et al., 2022). Unlike most regions that experienced economic contraction from COVID-19 through voluntary distancing, fear, or uneven mandates, Hawaii implemented one of the few deliberate and coordinated state-level shutdowns in the United States. This policy was aimed not merely at mitigating the virus, but at suppressing travel and commerce altogether. On March 26, 2020, the state imposed a mandatory 14-day quarantine on all arrivals, effectively sealing its borders.

This paper addresses a core policy question: What are the economic costs of decisive early action in the face of an uncertain crisis? While much has been written about the epidemiological and social effects of COVID-19 interventions, less is known about their economic toll. This is especially true in cases where policymakers opted for maximal containment. The most famous case was Wuhan's January 2020 lockdown, as studied by Ke and Hsiao (2022). However, work outside of this context is rare. Understanding these costs is essential not just for pandemic preparedness, but for future crises, such as climate shocks or geopolitical disruptions, where early intervention may again carry significant short-run economic risks in exchange for long-term societal benefits (Newman and Noy, 2023).

Estimating that cost, however, runs into an identification problem that is easy to overlook and fatal if ignored. Hawaii closed its border at the same instant the pandemic collapsed global travel demand. For a single treated unit these two forces—the *policy* and the *pandemic*—are perfectly collinear in time: every post-March-2020 month is simultaneously a "quarantine" month and a "pandemic" month. This is a coincident common shock, and it breaks the standard causal toolkit. A classical synthetic control or difference-in-differences design matches Hawaii to a counterfactual and reads a good pre-treatment fit as evidence of credibility. But any counterfactual Hawaii—whether built from other states or from Hawaii's own history—is a "no-quarantine" world that is *also* a "no-pandemic" world, because the design has no way to hold the pandemic fixed while switching the policy off. The resulting estimate is the border-closure effect plus whatever pandemic-driven demand loss Hawaii would have suffered anyway, and pre-fit quality says nothing about the size of that contamination.

The contribution of this paper is to take that identification problem seriously and to resolve it with the right tool. The confounder here—the pandemic—is a time-varying latent factor that moves the outcome and coincides with treatment timing; it cannot be conditioned away (it is unobserved) and it cannot be matched away (it moves any donor too). The *proximal* answer (Miao et al., 2018; Tchetgen Tchetgen et al., 2024) is not to remove the confounder but to instrument it: find observable negative controls that are driven by the same latent pandemic factor but carry no causal path from the treatment, and use them to algebraically subtract the pandemic out of the counterfactual. In this setting the negative controls are insulated employment sectors—industries hit by the pandemic recession but not downstream of whether tourists can physically arrive in Hawaii. I implement this through Single Proxy Synthetic Control (SPSC) (Park and Tchetgen Tchetgen, 2025), the member of the proximal family suited to a single treated unit with a single proxy group.

Empirically, contrasting SPSC against a naive Synthetic Historical Control (SHC) baseline (Chen et al., 2024) makes the identification point concrete: the gap between the two estimates is the pandemic's macro-recession component that the naive design silently loads onto the policy. SPSC recovers a large and robust tourism-demand collapse, but I am careful about what it identifies. Because the insulated sectors span the pandemic's general-recession factor and not its travel-demand-collapse factor, the SPSC tourism estimates over-attribute the drop to the policy and are best read as *upper bounds in magnitude* on the border-closure effect. I therefore report them as a robustness band rather than as point estimates and discipline each effect with a relevance diagnostic. Converting a bound into an actual point requires a proxy that carries the missing travel-demand factor, and I take that step with an external no-lockdown travel-demand donor—inbound air travel to Florida—which pulls the Visitor Days estimate off its upper bound toward a policy-centered point.

The remainder of the paper proceeds as follows. Section 1.2 reviews prior research on early pandemic interventions and the methodological challenge of studying one-off, jurisdiction-wide policies. Section 1.4 states the coincident-shock identification problem, the naive baseline, and the SPSC estimator. Section 1.5 describes the data. Section 1.6 reports the naive-versus-proximal contrast, the SPSC estimates and their identification diagnostics, and the robustness band. Section 1.7 discusses implications for public policy and crisis response.

1.2 Background

1.2.1 Tourism in Hawaii

Hawaii's economy transitioned from a plantation-dominated model in the late 19th and early 20th centuries to tourism as the primary driver after statehood. The sugar and pineapple industries, controlled by a few powerful firms known as the "Big Five," peaked in the 1960s but declined due to global competition, labor costs, and diversification efforts. Statehood in 1959, combined with the introduction of jet air travel, spurred a tourism boom, with visitor numbers surging from 171,000 in 1958 to over 6.7 million by 1990 (Croix, 2021). This shift modernized the economy, increased population growth, and generated substantial revenue, but it also created dependency on external factors like air travel and international markets.

Tourism is central to Hawaii's fiscal health, generating \$17.75 billion in visitor spending in 2019 alone, which supported \$2.07 billion in state tax revenue and 216,000 jobs across direct, indirect, and induced sectors. On any given day in 2019, there were 249,000 visitors contributing \$48.6 million in spending. Key islands like O'ahu (22.4 million daily spending) and Maui (14.0 million) bear the brunt of this activity (Hawai'i Tourism Authority, 2019). Leisure and hospitality employment, a proxy for tourism-related jobs, hovered around 190,000-

200,000 annually pre-2020, representing about one-third of the state's workforce. This reliance amplifies risks from shocks, as seen in past recessions, but also underscores tourism's role in funding public services and infrastructure.

Hawaii implemented some of the earliest and most aggressive COVID-19 containment policies in the United States. On March 4, 2020, Governor David Ige issued an emergency proclamation officially recognizing the virus as a statewide threat (Maui Now Staff, 2020). Within weeks, the state began taking unprecedented measures alongside states like California (Friedson et al., 2021). On March 17, Governor Ige publicly urged all visitors to suspend their trips to Hawaii for 30 days, anticipating substantial economic damage to the state's tourism-dependent economy. *"We do know there will be significant impact,"* he stated, though no concrete estimates were available at the time (Nakaso, 2020). A sweeping stay-at-home order followed on March 25, banning non-essential activities, closing non-critical businesses, and effectively shutting down day-to-day life and travel across the islands (HNN Staff, 2020). These actions were implemented alongside a mandatory 14-day quarantine on all arrivals, passed the same week, which further intensified the shutdown and isolated the state from both domestic and international travel. Despite the unprecedented scope of these interventions, no formal quantitative analysis has yet estimated the short-run economic effects of Hawaii's approach. This paper provides such an assessment.

1.2.2 Literature Review

The onset of the COVID-19 pandemic spurred a global wave of NPIs, with governments acting swiftly to limit viral transmission. Much of the early research focused on the public health rationale behind these policies—particularly the value of acting early to prevent exponential growth in cases. Friedson et al. (2021), for instance, study California's shelter-in-place order and argue that its primary purpose was to delay the pandemic's peak, buying time for hospitals to prepare. They emphasize the importance of California's timing—implementing the policy while infection growth was still modest—as a key to its effectiveness.

Other studies reinforce the value of early, aggressive containment. Using SCM, Yang (2021) analyzes Wuhan's lockdown and estimates substantial reductions in population movement and COVID-19 transmission, emphasizing the relevance of their findings for international public health policy. Cerqueti et al. (2021), studying Italy's decision to close schools, stress that timing is critical when dealing with exponential contagion. They also note a political economy challenge: if containment is successful, the intervention may appear excessive in retrospect, raising political costs for early action. Bayat et al. (2020) offer a quantitative example, estimating that deaths in New York could have been reduced by over 80% had lockdown been enacted ten days earlier. Outside of China and Europe, similar insights emerge. Sobiech Pellegrini (2022) and Li and Shankar (2023) document similar patterns in Brazil and Texas, finding that premature reopenings by state governments led to sustained spikes in cases.

Collectively, these studies underscore a key lesson: early interventions can mitigate public health disasters, but often at economic cost. This tradeoff between health and the economy became central to both policymaking and public discourse. Osman et al. (2022) document how opponents of lockdowns, particularly in conservative circles, criticized them as economically inefficient and ineffective. Gibson and Sun (2023) evaluate statewide stay-at-home orders using synthetic control, focusing on new jobless claims as an economic indicator of interest. Lee and Yang (2022) examine the labor market impacts of COVID-19 in South Korea, highlighting the economic consequences of NPIs. Ke and Hsiao (2022) study the economic impact of Hubei's lockdown. They show that while Hubei's lockdown caused a sharp downturn in the short term, the effects were short-lived—suggesting that delay or inaction might have led to more lasting harm.

Other work has assessed broader welfare outcomes of interventions. Cole et al. (2020) show how the Wuhan lockdown reduced air pollution and associated mortality using augmented SCM. Oliu-Barton et al. (2022) study vaccine mandates in Europe and estimate that increased vaccine uptake not only averted deaths but also prevented GDP losses—suggesting that public health policy can have broader economic upside when well-designed. Despite the breadth of this literature, Hawaii's case remains notably absent. Yet it represents perhaps the clearest U.S. example of an early, and most aggressive intervention. This paper contributes by bringing Hawaii into the empirical record and by offering credible estimates of the short-run economic costs of a "zero-COVID' ' style border (Yuan, 2022).

While much has been learned about the effects of pandemic interventions, measuring their economic impacts remains challenging, and Hawaii pushes that challenge to an extreme. Most evaluations of anti-COVID policy rely on difference-in-differences (DiD) or synthetic control methods (SCM). Where cross-sectional variation in policy exists—some counties or states locking down while others do not—SCM and panel methods have been used productively (e.g., Yang, 2021; Ke and Hsiao, 2023; Gong et al., 2024). But these designs require unaffected or minimally affected units to serve as comparisons, and, as Callaway and Li (2023) and Bilgel (2022) note, that assumption collapses in a global pandemic where nearly every region was shocked at once. Every other U.S. state and comparable island economy faced its own pandemic disruption, so there is no untreated Hawaii to borrow from across space. To my knowledge the only study examining Hawaii's pandemic economy is Bond-Smith and Fuleky (2022), which provides valuable descriptive analysis but constructs no counterfactual.

One response to the missing-donor problem is to borrow strength across *time* rather than space. The Synthetic Historical Control (SHC) method of Chen et al. (2024) does exactly this, reconstructing the treated unit's counterfactual from its own pre-treatment trajectory. SHC is the natural naive baseline for Hawaii and I retain it as such. But borrowing across time does not, by itself, solve the deeper problem here. Whether the counterfactual is built from other states (classical SCM) or from Hawaii's own history (SHC), it is a world with *neither* the quarantine *nor* the pandemic; extrapolating it into 2020 and differencing against the observed series attributes the entire post-March collapse—pandemic and policy together—to the treatment. The issue is not the source of the donor information but the *coincidence* of two shocks in a single unit.

That is precisely the structure the *proximal* / *negative-control* literature was built to handle.

Negative controls—variables driven by an unmeasured confounder but shielded from the treatment—have long been used to detect confounding (Lipsitch et al., 2010); proximal causal inference turns them into an identification strategy, using proxies of a latent confounder to recover causal effects that ordinary adjustment cannot (Miao et al., 2018; Cui et al., 2024; Tchetgen Tchetgen et al., 2024; Shi et al., 2020). Park and Tchetgen Tchetgen (2025) adapt this machinery to the synthetic-control setting as Single Proxy Synthetic Control, and Liu et al. (2024) develop a related surrogate-based proximal estimator. I apply SPSC to Hawaii because it is the one design in the synthetic-control family that concedes the pandemic confounder is present and unobservable and instruments it, rather than matching or conditioning it away.

1.3 A Gentle Introduction to Proximal Inference

Before the formal development, an analogy for the reader who has not yet met proximal inference, because the idea is far simpler than the machinery that carries it.

Imagine you want to know whether a fan cools your room. You switch the fan on—but at that same moment someone cracks the window, and a cool evening breeze drifts in. A few minutes later the room is cooler. How much of that cooling was the fan, and how much was the breeze? From inside this one room you cannot say: the fan and the breeze came on together, so the drop in temperature is the two of them combined, and nothing you measure in this room alone will separate them.

That is Hawaii's problem exactly. The fan is the border closure, the policy whose effect we want to isolate. The breeze is the pandemic, a force that was cooling tourism no matter what the state did. The two arrived in the same month, so the collapse we observe is fan plus breeze, and the treated unit on its own cannot tell us how much is which. A spotless pre-period track record does not rescue us; it only confirms that the room behaved normally before the window was ever opened.

The obvious repairs fail, and instructively they fail in opposite directions. Reason from the room's own past—“it was 75 degrees, now it is 68, so the fan did seven”—and you have quietly charged the breeze's cooling to the fan: you overstate the fan. Compare your room instead against other rooms that also had fans of their own running, and now the fan sits on both sides of the comparison and cancels, so it looks as though the fan did almost nothing: you understate it. One story is blind to the breeze; the other cancels the fan away.

The way out is to find the breeze somewhere the fan never reached. Look at the other rooms in the house whose windows were open—the same evening air—but which ran no fan of their own. Their cooling is the breeze alone, with the fan stripped off, so you can use them to reconstruct how much your room would have cooled from the breeze, subtract that, and keep what is left as the fan. Two things make a room usable. It must genuinely feel the same outside air and drift up and down with your room over an ordinary day (its *relevance*), and it must run no fan of its own (its *exclusion*). A sealed interior closet is useless, because it never feels the breeze; a room running its own fan is worse than useless, because it cancels the very effect we are trying to measure.

One last piece: not every fanless room feels the same breeze. Most register only the general cool-down of the whole house, while a single large window on the windward side catches the specific gust off the water. In Hawaii the insulated employment sectors are the general-cool-down rooms—they feel the macro recession but not the collapse in travel—while inbound air travel to Florida, drenched in the travel collapse yet never locked down, is the windward window. The move that feels backward is the crux of the method: a place the breeze hammered looks like a terrible comparison for your room and is in fact the only good one, precisely because it felt the shock and skipped the policy.

This is also why the method needs only a single proxy. You might hope the room's own history could stand in for the missing breezy room; it cannot, and the reason is sharp. Until tonight every room simply tracked the house thermostat together, so you never had occasion to learn how strongly any one room cools when its own window opens. A model that fits the room's calm past to perfection still cannot tell you its response to an open window. Hollingsworth and Wing (2022) make this precise: a factor that lies dormant before treatment and only wakes up afterward leaves no fingerprint in the pre-period, so a perfect pre-treatment fit is not the same as a correct answer. The treated room's own past carries part of what the estimator needs; the fanless, breezy room supplies the rest.

To turn the picture into something we can check, Figure 1.1 runs the fan experiment in a world whose true answer we know because we set it. Its left panel traces each method's reconstructed counterfactual against the observed series (black) and the true no-policy path (green dashed); its right panel bars the four estimated policy effects against that truth. We build a synthetic economy stirred by two hidden forces—an ordinary cycle, the house thermostat, and a sharp collapse after month 60, the breeze—and we install a policy whose true effect we fix at -15 points. Then we put the same question to each method: do you recover -15? In the single draw shown, reasoning from the room's own history (red) returns about -52, blind to the breeze; comparing against contaminated, fan-running rooms (orange) returns about -7, the fan cancelled away; Single Proxy Synthetic Control on the internal rooms alone returns about -45, catching the general cool-down but missing the gust; and adding the Florida window (blue) recovers about -16 against a truth of -15. The two naive methods miss in opposite directions; the negative control lands on the answer. Averaged over many draws the four settle near -51, -7, -42, and -16, and the ordering never changes.

The sections that follow trade the fan for notation—the factor model that makes "the breeze wakes up" precise (Section~1.4.1), the estimator and the conditions it requires (Section~1.4.3), and the Florida donor in the real data (Section~1.6.4)—but the fan is the whole of it: two coincident causes you cannot separate from one room, and a fanless, breezy room that lets you separate them.

1.4 Methodology

We know what happened once Hawaii mandated two-week quarantines. We do not know what would have happened had it not. The difficulty, developed in Section~1.4.1, is that the event we want to switch off (the border policy) is perfectly aligned in time with an event we cannot switch off (the pandemic). Section~1.4.2 states the naive Synthetic Historical Control baseline and why it cannot separate the two. Section~1.4.3 develops the Single Proxy

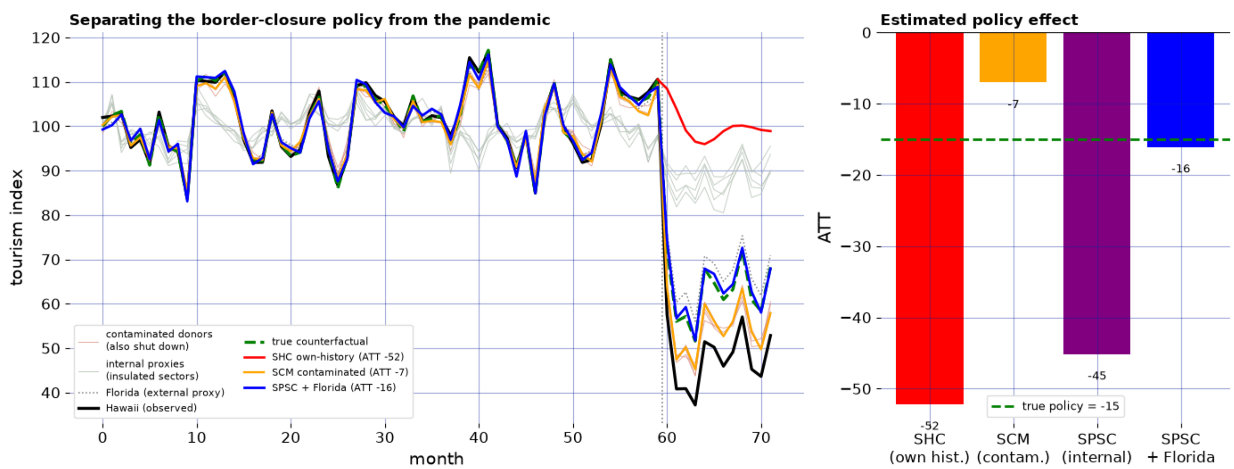


Figure 1.1: The fan experiment: four methods, one known answer.

Synthetic Control estimator that can, in principle, and Section~1.4.4 explains why the other obvious candidates do not.

Notation. Let $\mathbb{R}$ denote the reals and $\|\cdot\|_1$ the usual ℓ_1 norm. Calligraphic letters denote finite sets, with cardinality written $S = |\mathcal{S}|$; scalars are lowercase (h, t, n), vectors bold lowercase, and matrices bold uppercase $(\mathbf{W}, \mathbf{Y})$. Let $j \in \mathbb{N}$ index the N units, collected in $\mathcal{N}$, and $t \in \mathbb{N}$ index the T months, collected in $\mathcal{T} = \{1, \ldots, T\}$. Unit $j = 1$ is the treated unit (Hawaii); the remaining control units form $\mathcal{N}_0 := \mathcal{N} \setminus \{1\}$ of cardinality N_0. Treatment begins at $T_0 + 1$, with $\mathcal{T}_1 := \{t \leq T_0\}$ the pre-treatment period, $\mathcal{T}_2 := \{t > T_0\}$ the post-treatment period, and $n = T - T_0$ the number of post periods; $d_t = \mathbf{1}\{t > T_0\}$ is the treatment indicator. The observed outcome of unit j in month t is y_{jt}, its full path $\mathbf{y}_j = (y_{j1}, \ldots, y_{jT})^\top \in \mathbb{R}^T$; the treated path is $\mathbf{y}_1$, and the donor outcomes stack columnwise into $\mathbf{Y}_0 = [\mathbf{y}_j]_{j \in \mathcal{N}_0} \in \mathbb{R}^{T \times N_0}$. Write $\mathbf{W}_t := (y_{jt})^\top_{j \in \mathcal{N}_0} \in \mathbb{R}^{N_0}$ for the month-t cross-section of donor outcomes—the t-th row of $\mathbf{Y}_0$—which plays the role of the proxy vector below.

1.4.1 The identification problem: a coincident common shock

Following the Rubin potential-outcomes framework, let y^1_{1t} and y^0_{1t} be the treated and untreated potential outcomes for Hawaii (unit $j = 1$) at month t, with observed outcome $y_{1t} = d_t y^1_{1t} + (1 - d_t) y^0_{1t}$. The object of interest is the average treatment effect on the treated over the post period,

$$\text{ATT} := \frac{1}{n} \sum_{t \in \mathcal{T}_2} \left(y^1_{1t} - y^0_{1t} \right), \tag{1.1}$$

whose estimation requires a credible value for the unobserved y^0_{1t}—what Hawaii's economy would have done after March 2020 had it *not* closed its border.

To see precisely why the standard toolkit fails, place the problem in the interactive fixed-effects (factor) model that underlies modern synthetic control (Abadie et al., 2010; Bai, 2009; Ferman and Pinto, 2021; Zhang et al., 2022). With Hawaii as unit $j = 1$ and the candidate controls indexed by $j \in \mathcal{N}_0$, write

$$y^0_{1t} = \boldsymbol{\mu}_1^\top \boldsymbol{\lambda}_t + e_{1t}, \qquad y_{jt} = \boldsymbol{\mu}_j^\top \boldsymbol{\lambda}_t + e_{jt}, \quad j \in \mathcal{N}_0, \tag{1.2}$$

where $\boldsymbol{\lambda}_t \in \mathbb{R}^r$ is a vector of time-varying latent factors common to all units, $\boldsymbol{\mu}_j \in \mathbb{R}^r$ are unit-specific factor loadings, and $\mathbb{E}[e_{jt} \mid \boldsymbol{\lambda}_t] = 0$. In 2020 one coordinate of $\boldsymbol{\lambda}_t$ is the pandemic; it is (i) unobserved, (ii) loads on the outcome through $\boldsymbol{\mu}_1$, and (iii) switches on in the very month the policy does, so that d_t and the pandemic factor are collinear over $\mathcal{T}_2$.

A classical synthetic control seeks nonnegative weights $\boldsymbol{\gamma}^\dagger$ that reproduce the treated unit's loadings within the span of the donors',

$$\boldsymbol{\mu}_1 = \sum_{j \in \mathcal{N}_0} \gamma_j^{\dagger} \, \boldsymbol{\mu}_j, \tag{1.3}$$

the identifying condition of Abadie et al. (2010) and Ferman and Pinto (2021). When (1.3) holds, the synthetic control $\sum_{j \in \mathcal{N}_0} \gamma_j^{\dagger} y_{jt}$ is unbiased for y_{1t}^0 and the common factors—pandemic included—cancel in the post-period comparison. When it fails, differencing the observed series against any weighted counterfactual returns

$$y_{1t} - \sum_{j \in \mathcal{N}_0} \gamma_j y_{jt} = \underbrace{\left(y_{1t}^1 - y_{1t}^0\right)}_{\text{policy effect}} + \underbrace{\left(\boldsymbol{\mu}_1 - \textstyle\sum_{j \in \mathcal{N}_0} \gamma_j \boldsymbol{\mu}_j\right)^{\top} \boldsymbol{\lambda}_t}_{\text{factor-mismatch bias}} + \left(e_{1t} - \textstyle\sum_{j \in \mathcal{N}_0} \gamma_j e_{jt}\right), \tag{1.4}$$

so the estimate is the policy effect plus a bias equal to the residual loading mismatch projected onto the latent factors—dominated, over $\mathcal{T}_2$, by the pandemic coordinate. A good pre-treatment fit disciplines the first term but says nothing about the size of the second, because the pandemic factor is quiescent before March 2020 and only detonates after. This is the sense in which classical SC and difference-in-differences are naive here: they are unbiased only when the treated loadings lie in the donor span (1.3), and for a single unit hit by a coincident common shock that condition is neither testable nor, as Section~1.4.3 argues, satisfiable from tourism-exposed donors alone.

1.4.2 The naive baseline: Synthetic Historical Control

As the naive benchmark I use the Synthetic Historical Control (SHC) estimator of Chen et al. (2024), the strongest member of the class that ignores the pandemic factor, because it needs no external donors at all: rather than borrowing across units it borrows across time, reconstructing y_{1t}^0 from overlapping length-m windows of Hawaii's own smoothed pre-treatment history.[1]

In the factor language of (1.2), SHC's counterfactual is $\widehat{y}_{1t}^0 = \boldsymbol{\mu}_1^\top \widetilde{\boldsymbol{\lambda}}_t$, where $\widetilde{\boldsymbol{\lambda}}_t$ is the extrapolation into $\mathcal{T}_2$ of the treated unit's own pre-2020 factor path—a business-as-usual 2020 in which the pandemic coordinate never fires. Its bias is therefore

$$\widehat{y}_{1t}^0 - y_{1t}^0 = \boldsymbol{\mu}_1^\top \Big(\widetilde{\boldsymbol{\lambda}}_t - \boldsymbol{\lambda}_t \Big), \qquad t \in \mathcal{T}_2, \tag{1.5}$$

the treated unit's loading on the entire realized pandemic factor. Because SHC extrapolates a smooth pre-period trend it removes none of the pandemic—macro recession and travel-demand collapse alike—so its counterfactual is a no-pandemic, no-policy world, its estimate carries the full common-shock contamination, and it over-attributes the collapse to the border closure. Unlike the span condition (1.3) of a donor-based control, which at least nets out whatever factors the donors share, SHC's own-history extrapolation nets out nothing: it is the extreme

[1] Concretely, SHC smooths $\mathbf{y}_{\text{pre}}$ by local linear regression, forms a donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times S}$ of $S = T_0 - m - n + 1$ contiguous smoothed segments, and matches the final length-m evaluation window with simplex-constrained, ridge-regularized weights $\mathbf{w}^* = \operatorname{argmin}_{\mathbf{w} \geq \mathbf{0},\, \mathbf{1}^\top \mathbf{w} = 1} \|\widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w}\|_2^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2$, selecting the number of segments by a BIC-type forward rule and quantifying uncertainty by distribution-free conformal prediction intervals. Classical cross-sectional SCM is not even available here: it solves $\mathbf{w}^* = \operatorname{argmin}_{\mathbf{w} \in \Delta^{N_0}} \|\mathbf{y}_1 - \mathbf{Y}_0 \mathbf{w}\|_2^2$ over a simplex of *unexposed* donor units $\mathbf{Y}_0$, and in a global pandemic no such $\mathbf{Y}_0$ exists.

case of (1.4) in which the counterfactual is built from a single unit whose loadings match the treated unit exactly but whose factor realization is the wrong one. It is precisely this over-attribution that the proximal design of Section~1.4.3 is built to correct, and the size of the correction is the empirical payoff of Section~1.6.

1.4.3 Single Proxy Synthetic Control

The proximal strategy does not try to remove the pandemic factor; it instruments it. A negative control (or proxy) is a variable driven by the same latent confounder $\boldsymbol{\lambda}_t$ as the outcome but with no causal path from the treatment—the panel analogue of an instrument's exclusion restriction: it sees the pandemic, it does not see the quarantine. Here the donor set $\mathcal{N}_0$ is the $N_0 = 5$ insulated employment sectors (Section~1.5), so the proxy vector $\mathbf{W}_t = (y_{jt})^\top_{j\in\mathcal{N}_0}$ collects their month-t outcomes; they fell in the pandemic recession, so they load on the macro coordinate of $\boldsymbol{\lambda}_t$, but they do not depend on whether visitors can physically arrive in Hawaii, so no arrow runs from d_t to $\mathbf{W}_t$.

Single Proxy Synthetic Control (Park and Tchetgen Tchetgen, 2025) rests on two conditions relating the donors to the treated unit's untreated outcome. The first is a relevance (proxy) condition—the donors are informative about y^0_{1t},

$$\operatorname{Cov}\Big(h^\star(\mathbf{W}_t),\, y^0_{1t}\Big) \neq 0 \quad \text{for some } h^\star : \mathbb{R}^{N_0} \to \mathbb{R}, \qquad t \in \mathcal{T}_1, \tag{1.6}$$

their Assumption~3.1, which permits irrelevant donors so long as the remainder carry y^0_{1t}. The second is the existence of a synthetic control bridge function $h^\star$ that is conditionally unbiased for the untreated outcome,

$$y_{1t}^{0} = \mathbb{E}\Big[\, h^{\star}(\mathbf{W}_t) \,\Big|\, y_{1t}^{0} \Big] \quad \text{almost surely}, \qquad t \in \mathcal{T}, \tag{1.7}$$

their Assumption~3.2 and the key identifying restriction. Restricting to a linear bridge $h^{\star}(\mathbf{W}_t) = \mathbf{W}_t^{\top}\boldsymbol{\gamma}^{\star}$, condition (1.7) becomes the reverse-regression / measurement-error model

$$\mathbf{W}_t^{\top}\boldsymbol{\gamma}^{\star} = y_{1t}^{0} + e_t, \qquad \mathbb{E}\Big[\, e_t \mid y_{1t}^{0} \Big] = 0, \tag{1.8}$$

in which the synthetic control is an error-prone measurement of the treated counterfactual, rather than the counterfactual an error-prone function of the donors. This reversal is the essential difference from the span condition (1.3), and it is why SPSC needs only a single class of proxies where earlier proximal synthetic control requires two (Park and Tchetgen Tchetgen, 2025; Liu et al., 2024): the treated unit's own history supplies the second.

Identification of $\boldsymbol{\gamma}^{\star}$ is then a first-stage argument. In (1.8) the residual $-e_t$ is correlated with the regressor $\mathbf{W}_t$ but orthogonal to y_{1t}, while y_{1t} is correlated with $\mathbf{W}_t$ under (1.6); the treated unit's own outcome thus serves as an instrument for the endogenous donors—the instrumental-variable regression that Park and Tchetgen Tchetgen (2025) make explicit (their equations~(16)–(17)), in which a user-specified function of the treated unit's own pre-treatment outcome instruments the donors. This delivers the estimable moment condition (their Theorem~3.1, which shows $\mathbb{E}[h^\star(\mathbf{W}_t) \mid y_{1t}] = y_{1t}$ over $\mathcal{T}_1$)

$$\mathbb{E}\Big[\,\boldsymbol{\phi}(y_{1t})\Big(y_{1t} - \mathbf{W}_t^\top \boldsymbol{\gamma}^\star\Big)\Big] = \mathbf{0}, \qquad t \in \mathcal{T}_1, \tag{1.9}$$

where $\boldsymbol{\phi} : \mathbb{R} \to \mathbb{R}^p$ is a user-chosen vector of functions of the treated outcome that plays the role of the instrument set (I use the identity, $\boldsymbol{\phi}(y) = y$, so $p = 1$). To guard against the nonstationarity of year-over-year growth, the treated and donor series are de-trended against a low-dimensional cubic B-spline basis $\mathbf{D}_t$ (the SPSC-DT variant used throughout), giving the augmented estimating function

$$\boldsymbol{\Psi}_t(\boldsymbol{\eta}, \boldsymbol{\gamma}) = \begin{pmatrix} \mathbf{D}_t\Big(y_{1t} - \mathbf{D}_t^\top \boldsymbol{\eta}\Big) \\ \mathbf{g}(t, y_{1t}; \boldsymbol{\eta})\Big(y_{1t} - \mathbf{W}_t^\top \boldsymbol{\gamma}\Big) \end{pmatrix}, \qquad \mathbb{E}[\boldsymbol{\Psi}_t(\boldsymbol{\eta}^\star, \boldsymbol{\gamma}^\star)] = \mathbf{0}, \tag{1.10}$$

where $\mathbf{D}_t^\top \boldsymbol{\eta}^\star$ absorbs the secular mean of y_{1t}^0 and $\mathbf{g}$ stacks the de-trending and instrument blocks. Because $\boldsymbol{\phi}$ may be lower-dimensional than the donor pool ($p < N_0$), the moment system need not point-identify $\boldsymbol{\gamma}^\star$ on its own, so Park and Tchetgen Tchetgen (2025) regularize with a ridge penalty; the estimator solves

$$\Big(\widehat{\boldsymbol{\eta}}, \widehat{\boldsymbol{\gamma}}_\varrho\Big) = \underset{\boldsymbol{\eta}, \boldsymbol{\gamma}}{\operatorname{argmin}} \; \Big\{\overline{\boldsymbol{\Psi}}(\boldsymbol{\eta}, \boldsymbol{\gamma})^\top \,\widehat{\boldsymbol{\Omega}}\, \overline{\boldsymbol{\Psi}}(\boldsymbol{\eta}, \boldsymbol{\gamma}) + \varrho \, \|\boldsymbol{\gamma}\|_2^2\Big\}, \qquad \overline{\boldsymbol{\Psi}} = \frac{1}{T_0} \sum_{t \in \mathcal{T}_1} \boldsymbol{\Psi}_t, \tag{1.11}$$

for a positive-definite weight matrix $\widehat{\boldsymbol{\Omega}}$ and penalty $\varrho > 0$. Under the identity weight this admits the closed form $\widehat{\boldsymbol{\gamma}}_\varrho = (\widehat{\mathbf{G}}_{yW}^\top \widehat{\mathbf{G}}_{yW} + \varrho \mathbf{I})^{-1} \widehat{\mathbf{G}}_{yW}^\top \widehat{\mathbf{G}}_{yy}$ in the empirical cross-moments $\widehat{\mathbf{G}}_{yW}$ and $\widehat{\mathbf{G}}_{yy}$, the ridge stabilizing the solution when the donors are collinear. Given $\widehat{\boldsymbol{\gamma}} := \widehat{\boldsymbol{\gamma}}_\varrho$, Theorem~3.2 of Park and Tchetgen Tchetgen (2025) identifies $\mathbb{E}[y_{1t}^0] = \mathbb{E}[h^\star(\mathbf{W}_t)]$, so the counterfactual, effect path, and ATT are

$$\widehat{y}_{1t}^0 = \mathbf{W}_t^\top \widehat{\boldsymbol{\gamma}}, \qquad \widehat{\alpha}_t = y_{1t} - \widehat{y}_{1t}^0 \;\; (t \in \mathcal{T}_2), \qquad \widehat{\mathrm{ATT}} = \frac{1}{n} \sum_{t \in \mathcal{T}_2} \widehat{\alpha}_t, \tag{1.12}$$

recovered as the deterministic component of the post-period regression $y_{1t} - \widehat{y}_{1t}^0 = \mathrm{ATT} + \epsilon_t$. Inference uses the GMM sandwich standard error for $\widehat{\mathrm{ATT}}$ and, as a finite-sample cross-check that does not require a long post-period, the conformal prediction band of Park and Tchetgen Tchetgen (2025).

Relevance, and what the insulated sectors can span. The identifying content of SPSC lives in the relevance condition (1.6): the donors must genuinely track the treated unit's untreated outcome, exactly as a first stage must be strong. Its empirical footprint is the pre-treatment fit of the bridge, which I summarize by the pre-treatment root-mean-squared error $\mathrm{RMSE}_{\mathrm{pre}} = \{T_0^{-1} \sum_{t \in \mathcal{T}_1} (y_{1t} - \widehat{y}_{1t}^0)^2\}^{1/2}$; a pre-treatment RMSE small relative to the eventual effect signals that the reconstruction is informative rather than an artifact. Whether relevance holds, though, is ultimately a statement about the factor structure of (1.2), and here it is only partial. The pandemic is not one factor but (at least) two: a macro-recession factor—the general contraction, on which the insulated sectors load—and a travel-demand-collapse factor—people ceasing to fly anywhere out of fear, independent of any Hawaii policy—on which the insulated sectors carry essentially zero loading while tourism loads heavily. In the notation of (1.2), the tourism loading $\boldsymbol{\mu}_1$ lies in the donor span (1.3) along the macro direction but not along the travel-demand direction. SPSC built on insulated sectors alone therefore reconstructs only the macro component of y_{1t}^0 and cannot subtract the travel-demand collapse, so its tourism ATT absorbs that collapse and is an upper bound in magnitude on the border-closure effect, with the observed drop equal to policy plus travel-demand loss. No outcome the insulated sectors span alone escapes this: every treated series here loads on the travel-demand factor the donors miss, so all of the insulated-only estimates are bounds, not points. Closing the gap—turning a bound into a point—needs an external travel-demand negative control that carries that factor while staying immune to the border policy: inbound passengers to a comparable destination exposed to the pandemic but not locked down, such as Florida (Section~1.6.4).

1.4.4 Why the alternatives fail

Several tempting designs were considered and rejected; each breaks on the coincident-shock structure of Section~1.4.1. *Classical synthetic control* matches the pandemic into the counterfactual and so confounds policy with pandemic (equation~(1.4)). *Synthetic instrumental variables* need an instrument that shifts treatment but not the outcome directly; the treatment here is a single deterministic policy switch coincident with the shock, and no such instrument exists. *Forecast-based structural-break tests* cannot distinguish a latent level shift that is coincident with treatment from the treatment effect itself, and would assign the whole shift to the policy. *Surrogate methods* (Liu et al., 2024) require a separable effect/confounder channel and post-treatment surrogates predictive of the effect; here the confounder and the effect arrive together and no clean surrogate is available. *Orthogonalized* *synthetic control* sharpens *inference* under a maintained identification but does not solve the *identification* of a confounder coincident with treatment—it operates at the wrong layer. Proximal / negative-control inference is almost uniquely suited to this problem because it is the one framework that concedes the confounder is present and unobservable and instruments it.

1.5 Data

I assemble a monthly panel for Hawaii spanning January 1991 to December 2020, joining two public sources on month. All series are expressed as annualized year-over-year (YoY) growth rates. This transformation mitigates nonstationary trends and lets treatment effects be read directly as percentage-point changes in growth.

The treated outcomes are nine tourism and labor series from Hawaii's Department of Business, Economic Development & Tourism (DBEDT) and the Federal Reserve Economic Data (FRED) system: `Visitor Arrivals`, `Visitor Days`, `Occupancy`, `Mean Daily Rate`, `Revenue per Available Room`, `Accommodation Emp`, `Total Leisure Emp`, `Unemp Rate`, and `LFP`. Every one of these is downstream of the border closure: it moves when the ability of visitors to arrive changes, so it can only ever be a treated series, never a control.

The negative-control donors are five insulated employment sectors drawn from the DBEDT seasonally-adjusted Current Employment Statistics, the monthly job count by industry (Hawai'i Department of Business, Economic Development and Tourism, 2025), expressed in YoY growth: `NatRes_Constr_Emp` (mining, logging and construction), `Wholesale_Emp` (wholesale trade), `Financial_Emp` (financial activities), `HealthCare_Emp` (health care and social assistance), and `Government_Emp`. These industries were hit by the pandemic's macro-recession shock—falling roughly 7–12% in the 2020 trough—but do not depend on whether tourists can physically enter Hawaii, so no causal path runs from the quarantine to them. They are the empirical negative controls of Section~1.4.3.[2] Figure 1.2 contrasts the two groups: the treated tourism series collapse by 50–99% at the border closure while the insulated donors dip only modestly, the visual signature of a factor structure in which the donors carry the macro shock and the tourism series carry the macro shock plus the travel-demand collapse plus the policy.

The treatment indicator `Mandatory Quarantine` switches to 1 in March 2020, giving 10 post-treatment months (March–December 2020). Although the panel ships the full 1991–2020 history, the analysis truncates the pre-period to a recent window beginning January 2014 (74 pre-treatment months). Three decades of stale comovement is less relevant to the 2020 counterfactual than the recent history, and the 2014 window empirically sharpens the pre-treatment fit and yields non-degenerate GMM standard errors that the full-history-plus-detrend combination did not. The full-panel version is available by removing the truncation

[2]The five are the most insulated supersectors in the CES taxonomy (Hawai'i Department of Business, Economic Development and Tourism, 2025). I screened the full taxonomy for the negative-control property and deliberately *excluded* the sectors that are downstream of visitor arrivals and would therefore violate it: retail trade (visitor spending), transportation, warehousing and utilities (which bundles air transportation), arts, entertainment and recreation, accommodation and food services, and real estate, rental and leasing (car and vacation rental). Empirically these fell 23–62% in the 2020 trough, tracking the tourism collapse rather than the macro recession, which is exactly the contamination that disqualifies them as controls. A second tier of genuinely insulated sectors—manufacturing, professional and business services, and information—could enlarge the donor pool, but their pre-2020 comovement with the tourism series is no stronger than the five retained donors' ($|\text{corr}| \leq 0.4$), so they do not sharpen identification of the tourism (travel-demand) factor. The one donor that does carry that factor is external—inbound air travel to Florida, added in Section 1.6.4.

and enters the robustness band of Section~1.6.

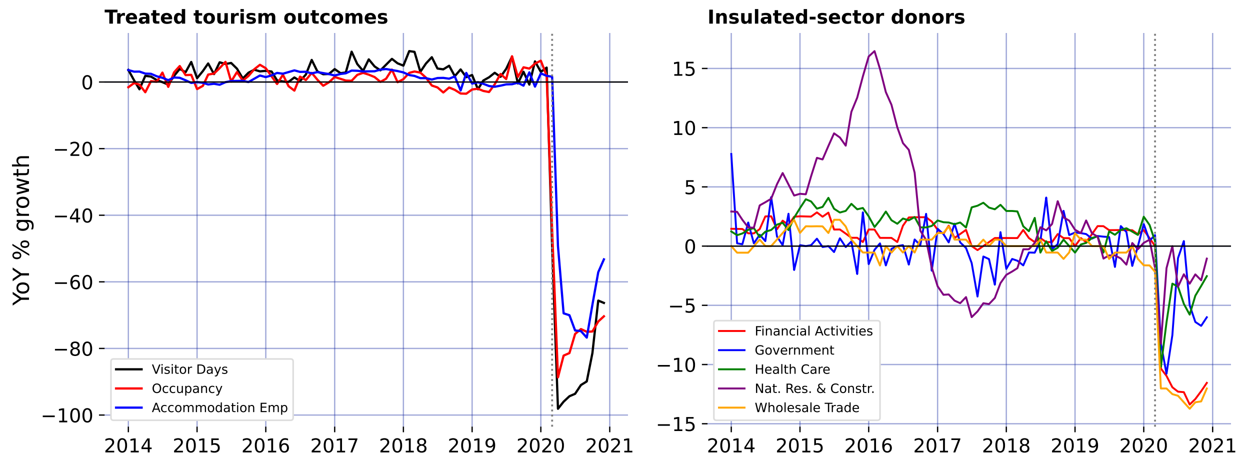


Figure 1.2: The negative-control premise. Left: treated tourism outcomes (year-over-year growth) collapse by 50–99% at the March 2020 border closure. Right: the five insulated-sector donors dip only modestly—they carry the pandemic's macro-recession factor but not the travel-demand collapse or the policy. The dotted line marks March 2020.

1.6 Results

1.6.1 The naive-versus-proximal contrast

The central methodological point is visible in a single comparison: the same treated series, estimated first with the naive Synthetic Historical Control baseline and then with proximal SPSC. Figure 1.3 and Figure 1.4 show it for Visitor Days. The naive SHC counterfactual (Figure 1.3) is a business-as-usual 2020 built from Hawaii's own smooth pre-pandemic history; differencing against it attributes essentially the entire post-March collapse to the policy, giving an ATT of -85.2 percentage points. The SPSC counterfactual (Figure 1.4) instead reconstructs Hawaii's untreated path from the insulated sectors, which *net out the pandemic's macro-recession component*, and returns a smaller ATT of -69.3 points.

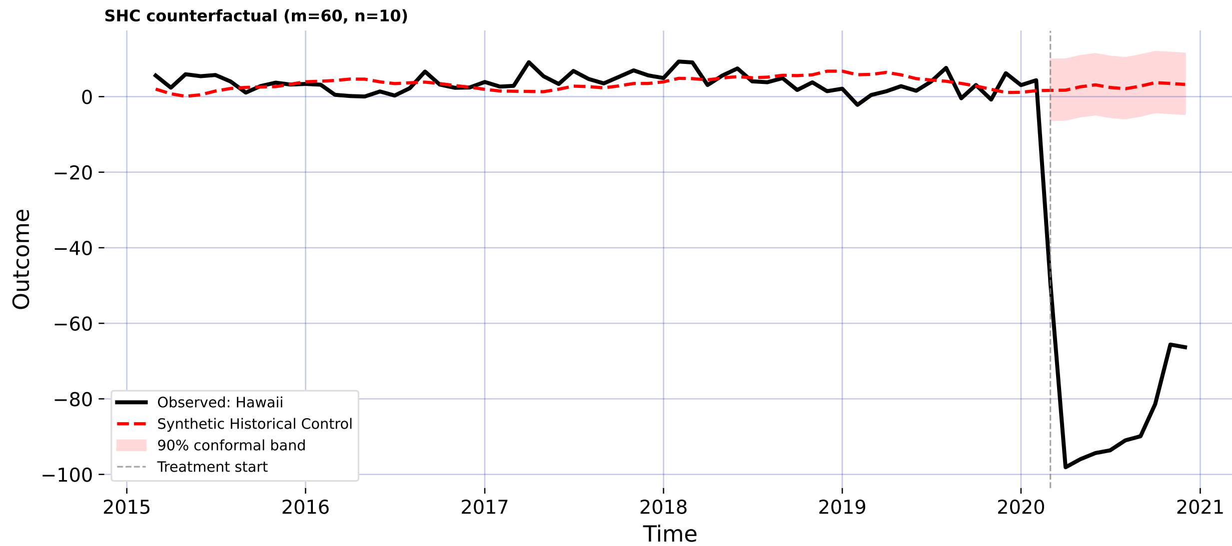


Figure 1.3: Naive baseline (Synthetic Historical Control), Visitor Days. The counterfactual is built from Hawaii's own pre-2020 history, so it is a world with neither the pandemic nor the policy and the full collapse loads onto the estimate.

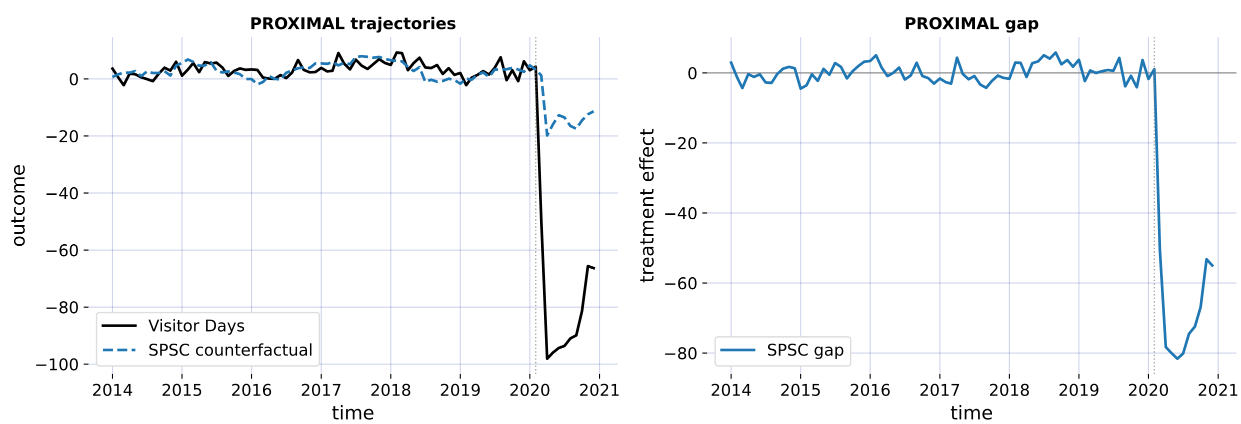


Figure 1.4: Proximal Single Proxy Synthetic Control, Visitor Days. The counterfactual (left panel) is reconstructed from the insulated-sector negative controls, netting out the pandemic's macro-recession factor; the gap panel (right) is the estimated effect path.

The 16-point gap between the two estimates is not noise: it is the factor-mismatch term of (1.4)—the pandemic's macro-recession contribution, which the naive design silently charges to the border policy and which SPSC removes by instrumenting it. The same wedge appears across the headline outcomes (Figure 1.5). It is largest, in level terms, for Revenue per Available Room, where the naive ATT of -84.0 points shrinks to -68.1 under SPSC—a 16-point pandemic component the insulated sectors net out because that series loads heavily on the general-recession factor they span.

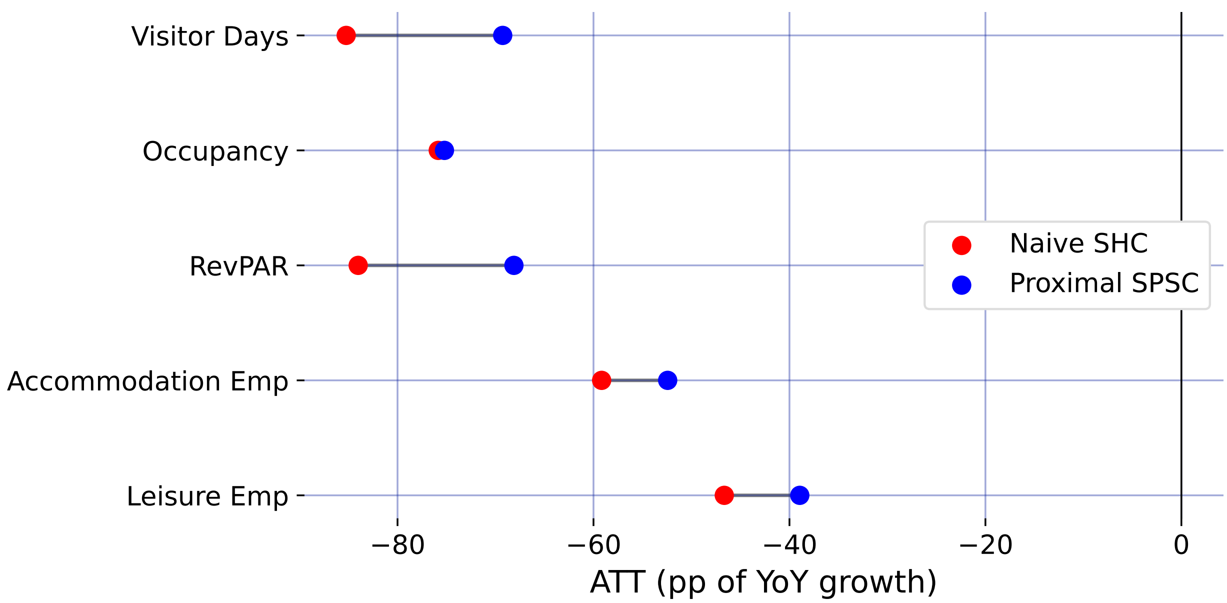


Figure 1.5: Naive SHC versus proximal SPSC average treatment effects (percentage points of year-over-year growth) for the headline outcomes. Each row pairs the naive Synthetic Historical Control estimate (red) with the proximal SPSC estimate (blue); the connecting segment is the gap—the pandemic's macro-recession component the naive design cannot net out.

1.6.2 SPSC estimates and the pre-treatment fit

Figure 1.6 reports SPSC-DT across all nine outcomes together with the pre-treatment RMSE of the bridge—the empirical footprint of the relevance condition (1.6). Reading the figure requires the discipline that diagnostic imposes: a point estimate is only as trustworthy as the pre-period fit of the synthetic control, and, as Section~1.4.3 argued, whether that fit reflects genuine relevance depends on which factors the insulated sectors can span.

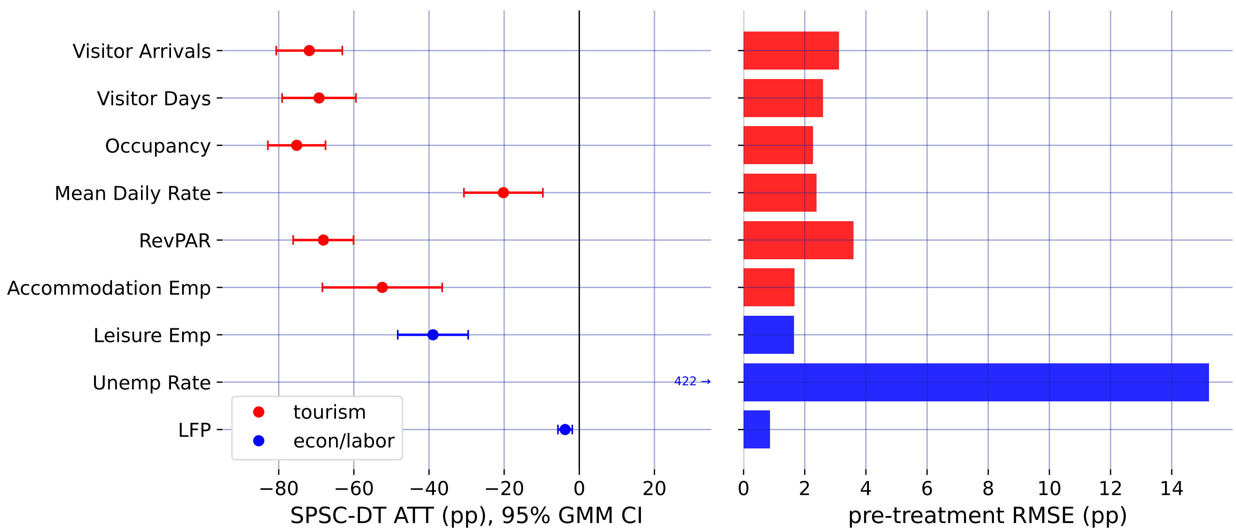


Figure 1.6: SPSC-DT estimates and pre-treatment fit, all nine outcomes (tourism red, econ/labor blue). Left: the ATT in percentage points of year-over-year growth, with the 95% GMM confidence interval (sandwich standard error) as the error bar. Right: the pre-treatment RMSE of the bridge—the root-mean-squared residual over the 74 pre-treatment months—the empirical footprint of the relevance condition.

Two readings follow. First, the tourism-demand collapse is large and robust: Visitor Days, Visitor Arrivals, Occupancy, and Revenue per Available Room all show a $\sim$ 65–75-point year-over-year drop against the proximal counterfactual, against pre-treatment RMSEs of only a few growth-rate points (for Visitor Days, 2.6 points against a $\sim$ 70-point effect)—the reconstruction is not an artifact of a poor fit. Second, the tourism estimates are nonetheless upper bounds in magnitude on the policy effect, not points: as Section~1.4.3 shows, the insulated sectors span the macro-recession factor but not the travel-demand-collapse factor, so the tourism loading $\boldsymbol{\mu}_1$ lies outside the donor span (1.3) along the travel-demand direction and the ATT still carries that component. The observed drop is policy plus travel-demand loss. No outcome the insulated sectors span alone delivers a clean point estimate—which is why the headline effects are reported as the band below, and why closing the gap to a point needs the external travel-demand donor of Section~1.6.4.

Two caveats attach to the GMM standard errors. On this 2014 window they are non-degenerate and usable for the visitor series ($\sim$ 3–8 points), a genuine improvement over the full-history window where de-trending a near-interpolating fit collapses the sandwich toward zero. But two of the labor series remain unreliable—the Unemployment Rate ATT is a relative-growth artifact with a near-zero SE, and the Labor Force Participation SE is tiny—and should not be read at face value. More fundamentally, even a well-behaved SE captures only *sampling* noise around the pre-period bridge, not the *extrapolation* risk of transporting that bridge across the March-2020 structural break, which is the dominant uncertainty here. I therefore do not lean on the SE alone.

1.6.3 Robustness: a specification band, not a point

Because the honest uncertainty is extrapolation risk rather than sampling noise, I report the headline effects as a band built from specification sensitivity: de-trending on versus off, the B-spline degrees of freedom, the polynomial basis degree, dropping each of the two largest-weight donors, and the full-1991 versus 2014 window. Figure 1.7 collects the minimum, median, and maximum SPSC ATT across these specifications for the three best-behaved outcomes.

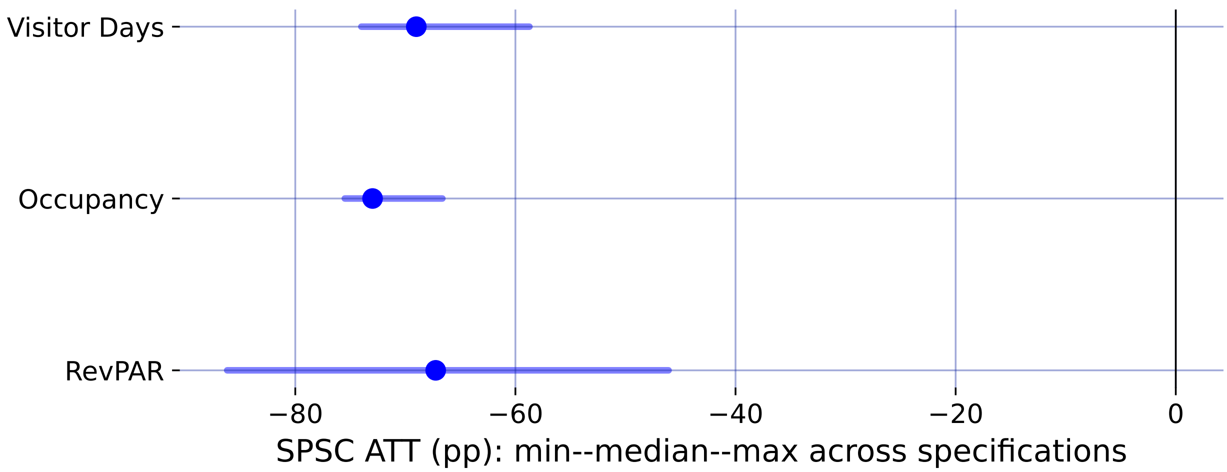


Figure 1.7: Specification robustness band. The point is the median SPSC ATT (percentage points) and the bar spans the minimum-to-maximum ATT across seven specifications: detrend on/off, spline df, basis degree, two donor-subset drops, and the 2014 versus full-1991 window.

For the visitor-volume series the band runs roughly from -74 to -59 points (median -69), so the defensible statement is a $\sim$ 60–75-point *upper-bound* collapse in tourism demand rather than any single figure. Taken together, the diagnostics and the band say the same thing: the border closure sits on top of a large tourism-demand collapse that the insulated sectors cannot fully separate from it, so the tourism numbers bound the policy effect from above. Recovering an actual point estimate—rather than a bound—requires spanning the missing travel-demand factor from outside the treated economy, which is the task of the next section.

1.6.4 An ensemble of exposed-but-open destinations

The insulated sectors span the macro-recession factor but not the travel-demand-collapse factor (Section~1.4.3), which is why the tourism estimates are bounds. Closing that gap requires a proxy that carries the travel-demand factor while remaining immune to Hawaii's border policy. Such a proxy is an *exposed-but-open* destination: a place whose inbound air travel collapsed with the pandemic yet that imposed no traveler quarantine, so it registers the travel-demand shock without registering the Hawaii policy. Rather than lean on one, I assemble a small ensemble of them from the Bureau of Transportation Statistics on-time-performance record (Bureau of Transportation Statistics, 2024)—monthly arriving-flight counts in YoY growth: Florida statewide (the archetype, its air traffic down about 70% at the 2020 trough) and three leisure metros in states that likewise never locked their borders—Tampa, Orlando, and Phoenix. Each enters the long panel as its own donor unit (`group = travel-demand`), and—this matters—I fit them *one at a time*, passing the insulated five plus a single open-destination donor, not all at once (I return to why below).

Figure 1.8 reports the convergence, and the headline is robustness through agreement. For Visitor Days, every one of the four independent open-destination donors pulls the estimate off the insulated-only bound of -69.3 points into a policy-centered band of roughly -51 to -65 points, and each one *improves* the pre-treatment fit rather than degrading it. That four proxies chosen independently—different cities, two different states—move the estimate the same way and by a similar amount is far stronger evidence that the observed drop is policy plus a real, separable travel-demand collapse than any single proxy could be. Florida gives the cleanest single fit (its Visitor Days pre-treatment RMSE falls from 2.59 to 1.93, bridge weight $\gamma_{\mathrm{FL}} = 0.15$), consistent with its pre-2020 correlation of +0.35 with Hawaii visitor volume.

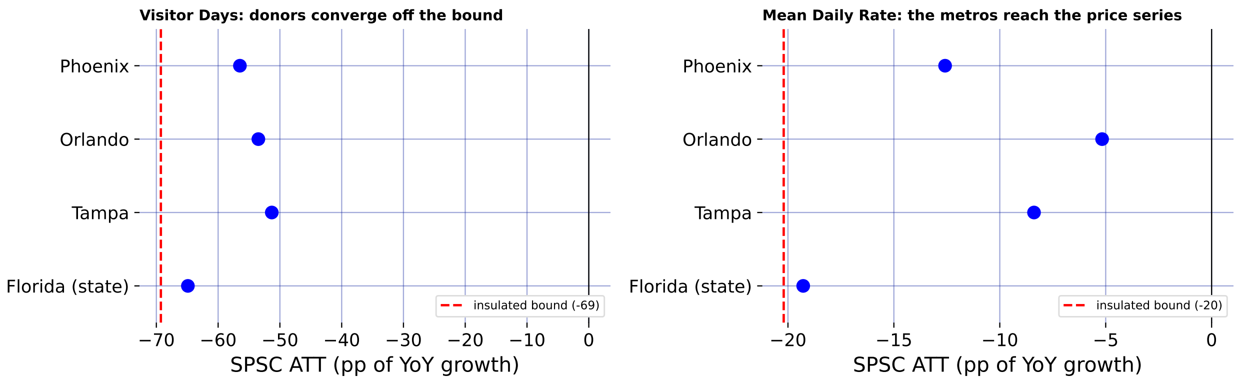


Figure 1.8: The open-destination ensemble. Each exposed-but-open donor (blue points), fit one at a time alongside the insulated five, is compared against the insulated-only estimate (red dashed line). Left, Visitor Days: every donor pulls the estimate off the upper bound toward a policy-centered value. Right, Mean Daily Rate: Florida (a volume series) leaves the price series untouched, but Phoenix—whose resort room rates track leisure demand—reaches it. The convergence across independent donors, not any single point, is the object of interest.

The ensemble also reaches an outcome a single volume proxy could not. Florida is a flight-*volume* series, so it comoves with Hawaii *visitor volume* but not with hotel prices, and leaves Mean Daily Rate essentially at its insulated-only value of -20.2 points. Phoenix—a desert-resort market whose room rates swing with leisure demand—does comove with Hawaii's daily room rate (pre-2020 correlation +0.36), and adding it pulls Mean Daily Rate from -20.2 toward -12.6 points while *improving* the fit (pre-treatment RMSE 2.37 to 1.81). Different open destinations carry the travel-demand factor as it manifests in different outcomes—volume for the Florida metros, price for Phoenix—so a small, well-chosen ensemble reaches further into the tourism block than any one proxy.

The discipline the design demands is the reason for fitting the donors *singly*. Added *en masse*—all four open destinations at once—the near-collinear proxies induce large offsetting ridge weights that *degrade* the fit exactly where a well-chosen single donor helped most, the price and utilization series (for Revenue per Available Room, the pre-treatment RMSE rises from 3.59 insulated-only to 4.39 with the full ensemble), the failure the relevance condition warns of. The lesson is not that more proxies are better but that more *independent* proxies, each read on its own, corroborate: the object is a convergence band, not a single super-donor.

Figure 1.9 shows the counterfactuals against the observed series for every headline outcome, using Florida as the anchor donor. The wedge between the insulated and the +Florida counterfactual is where the travel-demand donor does work—visible for the visitor-volume series, invisible for Occupancy, where Florida is (correctly) down-weighted to zero. Reading any single wedge as a clean travel-demand decomposition would over-interpret it, though: each +donor fit re-solves all bridge weights jointly, so the honest object is the convergence across donors in Figure 1.8, not one panel here.

1.7 Discussion

The methodological lesson of Hawaii is that a good pre-treatment fit is not identification when a policy and a global shock arrive in the same month. A naive synthetic control—whether it borrows across states or, as in SHC, across the treated unit's own history—produces a clean-looking counterfactual and a large ATT, but that ATT is the border-closure effect plus the pandemic's contribution to Hawaii's outcomes, and nothing in the fit reveals the mix.

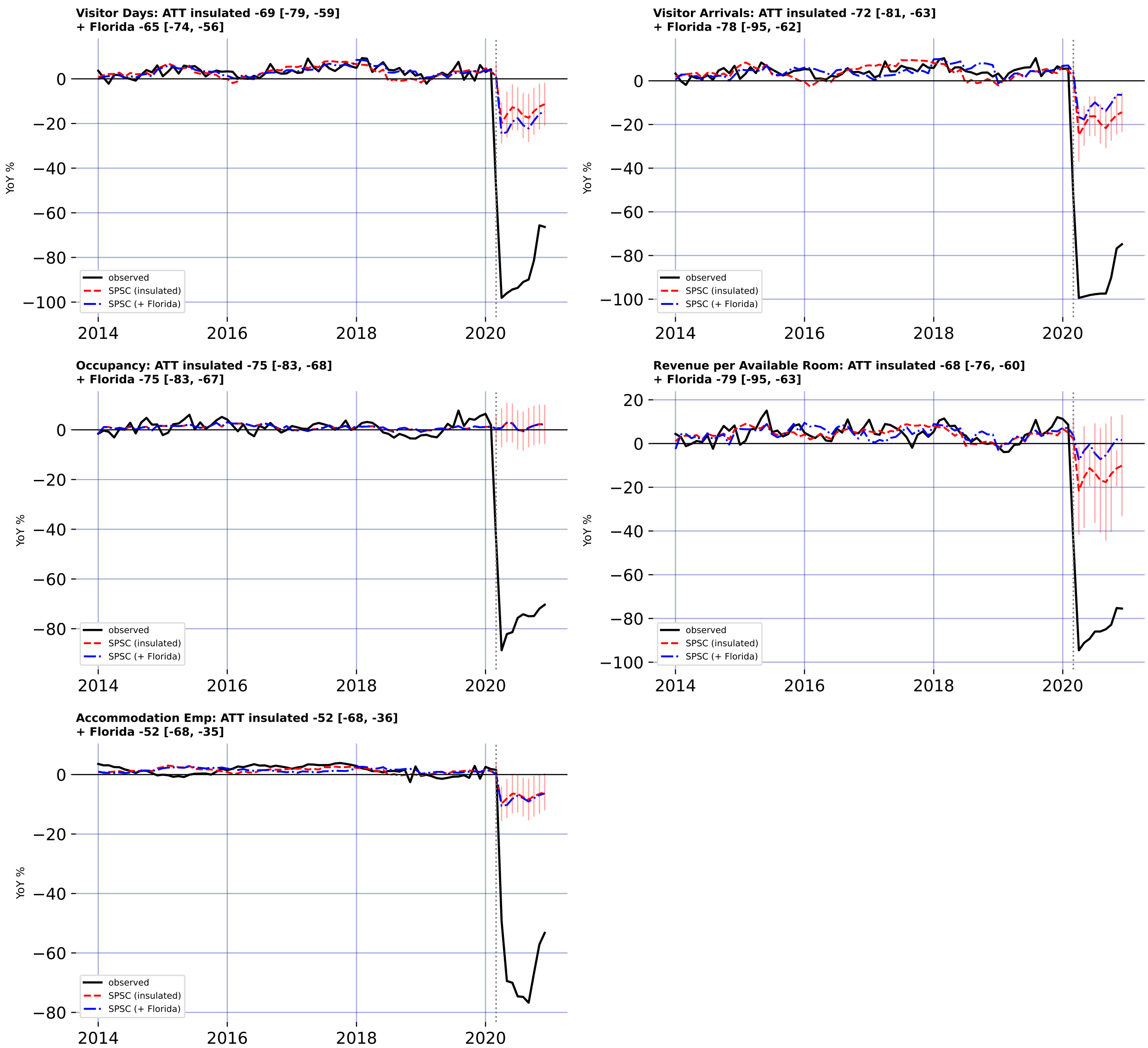


Figure 1.9: SPSC counterfactuals for the five headline outcomes: observed (black), the insulated-donor counterfactual (red dashed), and the counterfactual after adding the Florida travel-demand donor (blue dash-dot). Red spikes are the per-period conformal prediction interval around the insulated counterfactual (clipped to each panel; wide/degenerate for the near-interpolating series, where the ATT interval is the informative uncertainty). Each panel title gives the ATT with its 95% GMM confidence interval for both donor sets.

Proximal / negative-control inference is the discipline that separates them: by instrumenting the latent pandemic factor with insulated employment sectors that carry it but are immune to the policy, SPSC nets out the macro-recession component the naive design cannot. The wedge between the two estimates—about 16 points for Visitor Days, and larger still for the most recession-exposed of the headline series—is a direct, quantitative measure of how much pandemic contamination a naive reading would have mistaken for policy.

At the same time, the analysis is candid about the limit of what the shipped data can identify. The insulated sectors span the pandemic's general-recession factor but not its travel-demand-collapse factor: even with an open border, Hawaii tourism would have fallen sharply as people stopped flying. SPSC on insulated sectors alone therefore recovers only the macro component of the counterfactual, and the resulting tourism ATTs over-attribute the collapse to the policy—they are upper bounds in magnitude, which is why I report them as a band rather than as points. Recovering an actual point estimate from insulated donors alone is not possible here, because every treated series loads on the travel-demand factor those donors miss; the point has to be spanned from outside the treated economy, which is what the open-destination donors do. The substantive conclusion is thus deliberately modest and, I would argue, more credible for it: the border closure coincided with a near-total collapse in measured tourism demand, of which a bounded share is causally the policy, with a convergent ensemble of external travel-demand donors pulling the Visitor Days estimate off its upper bound toward a policy-centered band.

Section~1.6.4 takes the decisive step of closing the two-factor gap with external travel-demand negative controls—inbound air travel to Florida and to a set of other exposed-but-open leisure markets that did *not* lock down—and the result is instructive about both the promise and the discipline the approach demands. For Visitor Days the open-destination donors do what the design promises, and they do it *in agreement*: fit one at a time, all four independently pull the estimate off the insulated-only bound into a policy-centered band, each improving the pre-treatment fit. Because the donors are chosen independently—different cities, two different states—their convergence is far stronger evidence that the missing factor is real and separable than any single proxy could be. A well-chosen ensemble also reaches further than one volume series: Phoenix, whose resort room rates track leisure demand, pulls the daily-room-rate series that Florida (a volume proxy) leaves essentially untouched. But the ensemble does not mechanically dissolve every bound, and the search must stay disciplined: near-collinear open destinations added *en masse* induce large offsetting weights and hurt the fit, so the donors corroborate as a convergence band rather than combine into a single super-donor; and Puerto Rico, tempting as a fellow island, is not a valid control because it imposed its own stay-at-home order in mid-March 2020 and is a contaminated, treated-like unit that, if anything, bounds the effect from the other side.

These estimates speak only to short-run economic outcomes through the end of 2020 and do not weigh the public-health benefits—lives saved, hospital strain averted—against the economic cost. What the paper does provide is a credibly bounded price tag for decisive early action, and a template for evaluating it: when the intervention and the crisis it responds to are one and the same event in time, the analyst's task is not to find a cleaner comparison unit but to find a variable that sees the crisis and not the policy.

1.8 Conclusion

This paper estimates the short-run economic consequences of Hawaii's March 2020 mandatory quarantine while confronting the identification problem that its predecessors on this topic could not: for a single treated unit, the border-closure policy and the COVID-19 pandemic are perfectly collinear in time, so no synthetic control that merely matches Hawaii to a pre-pandemic counterfactual can separate the two. I resolve this with Single Proxy Synthetic Control, a proximal / negative-control estimator that instruments the latent pandemic factor using insulated employment sectors that carry the pandemic's macro-recession shock but are immune to the border policy.

Contrasting SPSC with a naive Synthetic Historical Control baseline shows that a substantial share of the measured tourism collapse—roughly 16 percentage points for Visitor Days—is pandemic contamination the naive design charges to the policy. SPSC recovers a large, robust tourism-demand collapse (about 60–75 points of year-over-year growth for the visitor-volume series) but, because the insulated sectors span only the macro-recession factor and not the travel-demand-collapse factor, they under-span the tourism factor and identify these effects only weakly; the estimates from insulated donors alone are upper bounds in magnitude on the policy effect and are reported as a robustness band rather than as points. Because every treated series loads on the travel-demand factor those donors miss, none of the insulated-only estimates is a clean point; the trustworthy point comes only once the missing factor is spanned from outside the treated economy, which is the role of the external Florida travel-demand donor.

Methodologically, the study demonstrates how negative-control inference disciplines causal claims when a policy and a global shock coincide—a structure common to pandemic border closures, but also to climate shocks and other systemic crises where a would-be counterfactual unit is itself engulfed by the event. Substantively, it delivers a bounded, honest estimate of the economic price of Hawaii's "Great Isolation with Aloha characteristics''. Adding a small ensemble of external no-lockdown travel-demand donors—inbound air travel to Florida and to the Tampa, Orlando, and Phoenix leisure markets—takes the step from a bound to a policy-centered band: fit independently, they converge in pulling the Visitor Days effect toward the policy-only magnitude, and one of them (Phoenix) reaches the room-rate series a single volume proxy cannot. Finishing the job for every outcome is a matter of assembling more such open-destination proxies, disciplined by relevance, rather than a limit of the design.

CHAPTER 2

Clearing the Air: How India's 2020 Lockdown Impacted Air Quality

2.1 Introduction

The COVID-19 pandemic prompted governments worldwide to implement non-pharmaceutical interventions (NPIs), such as mass lockdowns, to curb virus spread and protect public health systems. India's nationwide lockdown, beginning on March 25, 2020, was among the strictest globally since the Wuhan lockdown in China, halting mobility, industry, and construction for 1.3 billion people. Early research focused on virus outcomes (Yang, 2021; Bayat et al., 2020) and economic impacts (Gong et al., 2024; Wu et al., 2023), but lockdowns also likely influenced air quality, a critical public health factor. Popular media reported improved air quality during lockdowns, later confirmed by studies documenting reductions in pollutants such as PM2.5 (Watts and Kommenda, 2020; Venter et al., 2020; Kumari and Toshniwal, 2021). Given that India has some of the most polluted cities in the world, understanding how large-scale interventions affect pollution levels is of great interest. However, many studies are descriptive and do not estimate counterfactual scenarios—what air quality would have been without lockdowns—limiting their policy relevance, or rely on strict assumptions for identification (Milman and Uteuova, 2025).

This study quantifies the causal impact of India's 2020 national lockdown on PM2.5 levels nationally and in major cities using the Synthetic Historical Control (SHC) method (Chen et al., 2024). Unlike studies that rely on untreated comparison groups—which are unviable in the COVID-19 context—or on covariate conditioning, I develop an extension of SHC that accommodates imperfect pre-treatment fit. By constructing counterfactuals from historical data with flexible, non-parametric methods, SHC isolates the lockdown's effects without requiring unexposed control units. In addition to the national analysis, this study examines Delhi, Mumbai, Bangalore, and Kolkata—megacities consistently among the world's most polluted, and major hubs of finance and commerce. Embedding these cases in a unified causal framework moves beyond simple pre–post comparisons and addresses a key policy question: what happens to air pollution when a country of India's scale implements a nationwide shutdown? These findings can inform sustainable air quality management in India and provide lessons for urban planning and climate change mitigation globally.

The rest of the paper is organized as follows. Section 2.2.1 provides background on air pollution in India, highlighting its health and economic costs. Section 2.2.2 summarizes existing literature and methodological gaps. Section 2.2.2.2 outlines the empirical strategy, including SHC and the augmented version proposed here. Section 2.3 details data collection and processing.

2.2 Background and Previous Work

2.2.1 Policy Context

Air pollution is one of India's most pressing environmental and public health challenges. This section provides context on pollutant sources, exposure levels, health and economic impacts, and how the COVID-19 lockdown created a natural experiment for studying air quality.

2.2.1.1 Sources and Severity of Air Pollution

Fine particulate matter (PM2.5), which penetrates deep into the lungs and bloodstream, is a primary pollutant of concern. Major contributors include vehicular emissions, industrial activities, construction dust, biomass burning for cooking and heating, and agricultural residue burning, particularly in northern states during winter. Natural factors such as dust storms and seasonal weather patterns exacerbate pollution levels. Rapid urbanization and population growth have intensified exposure, affecting over 1.4 billion people.

Recent data highlight the severity: in 2024, India's average annual PM2.5 concentration was 50.6 $\mu g/m^3$, far exceeding the WHO guideline of 5 $\mu g/m^3$ (Statista, 2025). Regional variation is significant—cities in the Indo-Gangetic Plain, like Delhi, often exceed 100 $\mu g/m^3$ during peak winter, while southern cities such as Chennai frequently surpass both WHO and national standards (Bhuyan et al., 2025; Singh et al., 2021).

2.2.1.2 Health Impacts

The health consequences of chronic PM2.5 exposure are substantial. Long-term exposure increases the risk of respiratory and cardiovascular diseases, including COPD, asthma, lung cancer, stroke, and heart disease (Joshi et al., 2025). In India, air pollution causes roughly 1.5 million premature deaths annually, with each 10 $\mu g/m^3$ rise in PM2.5 linked to an 8.6% increase in mortality (Jaganathan et al., 2024). Vulnerable populations, namely children, the elderly, and those with weaker immune systems, are most affected. Fine particulate pollution reduces average life expectancy by 5.3 years compared to WHO guidelines, with residents in

Delhi facing up to a 10-year reduction (BBC, 2022). Short-term PM2.5 spikes also increase daily death rates by 1.42–3.57% per 10 $\mu g/m^3$ (de Bont et al., 2024).

2.2.1.3 Economic Impacts

Air pollution imposes significant economic costs through healthcare expenditures, lost labor productivity, and impacts on sectors such as tourism and agriculture. Estimates suggest annual losses of USD 95 billion, roughly 1.36% of GDP (India State-Level Disease Burden Initiative Air Pollution Collaborators, 2020; Gupta and Damani, 2021). In the Delhi NCR, productivity declines and reduced tourist arrivals contribute substantially to these costs (Dalberg Advisors, 2021; Our Common Air Commission, 2024). Efforts like the National Clean Air Programme (NCAP), launched in 2019, aim to reduce PM2.5 and PM10 concentrations by 20–30% in 131 non-attainment cities by 2024–25 (Kawano et al., 2025).

2.2.1.4 The COVID-19 Lockdown as a Natural Experiment

The 2020 nationwide COVID-19 lockdown provides a unique opportunity to study rapid air quality improvements. Imposed on March 25, the lockdown halted transportation, industrial operations, and construction, leading to reductions in PM2.5 by 31–43%, with some cities experiencing drops up to 50% (Saleem et al., 2024; Nigam et al., 2021; Mishra et al., 2021). In megacities like Delhi and Mumbai, the Air Quality Index (AQI) improved by as much as 58%, primarily due to lower vehicular and industrial emissions (Zhang et al., 2020).

Although temporary, these reductions highlight the potential of targeted interventions. PM2.5 reductions were most pronounced in early lockdown phases, averaging 30–50% across monitored sites, with occasional spikes due to meteorology (Hu et al., 2022; Goel et al., 2021; Wang et al., 2020). Other pollutants, including NO2 and PM10, also decreased, consistent with global lockdown trends (Venter et al., 2020). Understanding these dynamics is crucial for causal evaluation and future policy planning.

2.2.1.5 Lockdown Policy Phases

India's lockdown, enacted under the Disaster Management Act of 2005, unfolded in multiple phases. Phase 1 (March 25–April 14) involved a complete nationwide shutdown, prohibiting non-essential movement, closing factories, and suspending public transport. Phase 2 (April 15 onwards) allowed a gradual reopening of essential services and agriculture, with color-coded zoning based on infection rates (Deshpande, 2022; Ranjan et al., 2020). While primarily a public health measure, the lockdown served as a large-scale quasi-experiment in emission control, revealing the disproportionate contribution of human activities to PM2.5 levels.

2.2.2 Literature Review

2.2.2.1 Methodological Aspects of Prior Literature

To contextualize our approach, we review existing studies on lockdown-related air quality changes and highlight key methodological gaps. Early studies primarily relied on descriptive pre–post comparisons or satellite imagery without causal adjustment (Mahato et al., 2020; Pathakoti et al., 2021; Srivastava et al., 2020; He et al., 2020; Wang et al., 2020). These approaches document reductions in pollutants but cannot answer counterfactual questions—what would air quality have been absent the lockdown? Some studies have applied causal models. For example, Heffernan et al. (2025) use causal machine learning and interrupted time series (ITS) methods to find reductions in NO2 levels. ITS was applied in Italy (Cameletti, 2020), finding no meaningful changes in air pollutants. Augmented Synthetic Control methods were used for Wuhan (Cole et al., 2020), showing a 63% reduction in NO2. Despite these examples, a recent systematic review finds that most lockdown–air pollution studies remain descriptive or non-causal (Silva et al., 2022), limiting their policy relevance.

2.2.2.2 Gaps and Contribution of This Study

A central gap is the lack of causal designs for national-scale interventions. Many studies use simple pre–post t-tests (Goel et al., 2021), graphical analyses (Kumari and Toshniwal, 2020; Benchrif et al., 2021), year-over-year comparisons (Le Quéré et al., 2020), or OLS regressions (Pal et al., 2022). These methods cannot fully account for confounding trends or temporal patterns, reducing confidence in policy-relevant estimates. SCM improves on this by constructing a counterfactual from untreated donor units as a weighted average (Ben-Michael et al., 2021a). However, SCM is limited when studying national policies, where no valid donor pool exists, given that SCM requires untreated control units in order to be viable. Similarly, causal machine learning approaches (Heffernan et al., 2025) rely on strong ignorability assumptions that may not hold for air pollution data with unmeasured year-specific factors. In contrast, the SHC method avoids these limitations by constructing counterfactuals directly from historical patterns, without assuming complete covariate measurement. This study extends SHC to an Augmented SHC (ASHC) framework, allowing limited extrapolation to improve pre-treatment fit while maintaining stability. By applying ASHC to India's national and city-level PM2.5 data, this work provides the first rigorous causal estimates of the 2020 lockdown's air quality impacts in a country of India's scale.

Methodology

Below I describe the Synthetic Historical Control (SHC) methodology from Chen et al. (2024). I then extend this framework to settings where pre-treatment fit is poor, yielding the Augmented Synthetic Historical Control (ASHC). The ASHCM is applied separately to the national average of PM2.5 for India, as well as Delhi, Mumbai, Bangalore, and Kolkata.

2.2.3 The Synthetic Historical Control Method

SHC estimates the causal effect of an intervention by constructing counterfactual outcomes from historical segments of the treated unit itself, rather than from contemporaneous control units. This makes it well-suited for national-scale interventions such as India's 2020 lockdown, where no valid donor pool of untreated countries exists. Before presenting the estimator, I state the assumptions that guarantee SHC's validity.

2.2.3.1 Assumptions

Assumption 1 (Error Properties)**.**

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathbb{E}[\varepsilon_t^2] < \infty \qquad \forall\, t \in \mathcal{T}. \tag{2.1}$$

This ensures that noise is well-behaved. The mean-zero condition rules out systematic bias, while the finite variance condition prevents extreme fluctuations.

Assumption 2 (Latent Smoothness and Matching).

$$\widetilde{y}(t) \in C^{H+1}, \quad H \geq 3, \qquad \sup_t \left| \frac{d^k}{dt^k} \widetilde{y}(t) \right| < \infty \;\; \forall\, k \leq H+1, \tag{2.2}$$

and there exists

$$\mathbf{w}^* \in \Delta^{N-1} \qquad \textit{such that} \qquad \mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}^*. \tag{2.3}$$

Smoothness ensures that the latent process can be approximated by a finite expansion. The matching condition guarantees that some convex combination of historical donor segments can reproduce the evaluation period. This parallels the "existence of weights" assumption in the SCM literature (Abadie et al., 2010; Amjad et al., 2018). Importantly, persistent factors such as meteorology are implicitly matched so long as they appear in the historical trajectory.

Assumption 3 (Feasibility and Stability).

$$\exists\, \mathbf{w} \in \Delta^{N-1} \;\; \textit{sparse, such that} \;\; \mathbf{y}_{eval} \approx \widetilde{\mathbf{Y}}_{pre}\mathbf{w}, \qquad \widetilde{\mathbf{Y}}_{pre}^{\top}\widetilde{\mathbf{Y}}_{pre} \succ 0. \tag{2.4}$$

Sparsity means that only a small number of donor segments are relevant. Positive definiteness prevents collinearity and ensures stable weights. In plain terms, SHC gives us a valid estimate of the lockdown's impact as long as three conditions hold. First, the random "noise" in pollution measurements behaves well — it averages out and does not swamp the real signal. Second, the history of pollution before the lockdown is smooth and informative enough that past patterns can be combined to mimic what would have happened in the absence of the lockdown. This means that unobserved influences like weather are implicitly accounted for, so long as they leave a stable footprint in the historical data. Third, the relevant pieces of history we draw on to form the counterfactual are not redundant, so the weights we place on them are stable rather than erratic. Under these conditions, SHC can reliably construct the missing counterfactual trajectory and deliver a valid estimate of the average treatment effect of the lockdown.

2.2.3.2 Estimator

Let $\mathcal{T} = \{1, \ldots, T\}$ index time, with pre-treatment period $\mathcal{T}_1 = \{t \in \mathcal{T} : t \leq T_0\}$ and post-treatment period $\mathcal{T}_2 = \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$, and an evaluation window of length m is defined as the final m pre-treatment periods $\mathcal{T}_{\text{eval}} = \{T_0 - m + 1, \ldots, T_0\}$. Let $\mathbf{y} = (y_1, \ldots, y_T)^\top$ denote the observed outcome series. The evaluation subvector is $\mathbf{y}_{\text{eval}} = \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector is $\mathbf{y}_{\text{post}} = \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$. For each $t \in \mathcal{T}$, let y_t^1 and y_t^0 denote the potential outcomes under treatment and no treatment, respectively. The observed outcome is

$$y_t = d_t y_t^1 + (1 - d_t) y_t^0, \qquad d_t = \mathbf{1}\{t > T_0\}. \tag{2.5}$$

For $t > T_0$, only y_t^1 is observed. SHC provides an estimate $\widehat{y}_t^0$ for the missing counterfactual, and the treatment effect is estimated as

$$\widehat{\alpha}_t = y_t - \widehat{y}_t^0, \qquad t \in \mathcal{T}_2, \tag{2.6}$$

with the average treatment effect on the treated (ATT) defined as

$$\widehat{\text{ATT}} = \frac{1}{n} \sum_{t \in \mathcal{T}_2} \widehat{\alpha}_t. \tag{2.7}$$

The implementation of SHC begins by smoothing the pre-treatment outcome series $\mathbf{y}_{\text{pre}}$ using local linear regression with a Gaussian kernel, yielding a smooth trajectory $\widetilde{\mathbf{y}}$ (Fan and Gijbels, 1992). From this smoothed series, we construct the donor matrix by extracting N overlapping segments, each of length m.

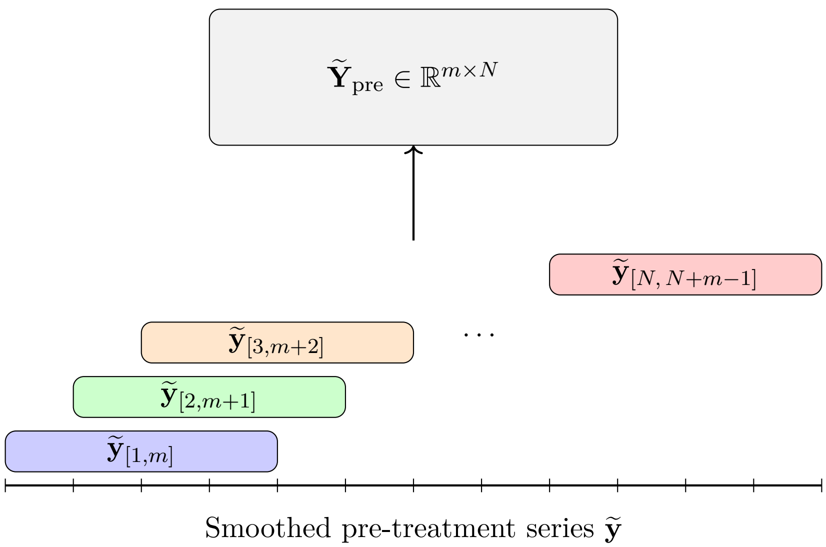


Figure 2.1: Constructing the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}}$ by sliding overlapping windows of length m across the smoothed pre-treatment series $\tilde{\mathbf{y}}$. Each colored segment corresponds to a column in the donor matrix.

The evaluation vector $\mathbf{y}_{\text{eval}}$ is reconstructed as a convex combination of the donor segments by solving the quadratic program

$$\mathbf{w}^* = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2, \qquad \text{s.t. } \mathbf{w} \geq 0, \; \mathbf{1}^\top \mathbf{w} = 1,$$

where $\mathbf{C}_0$ contains the eigenvectors of the donor Gram matrix associated with small eigenvalues. The second term penalizes directions of low variance, improving numerical stability when donor segments are collinear.

In practice, only a few donor segments contribute meaningfully to the reconstruction of $\mathbf{y}_{\text{eval}}$. To identify these, I use the forward selection algorithm described by Chen et al. (2024). Applying the resulting weight vector $\mathbf{w}^*$ to the aligned post-treatment donor matrix produces $\widehat{\mathbf{y}}^0_{\text{post}}$, the SHC estimate of the untreated counterfactual trajectory.

A further advantage of SHC in this setting is that it naturally accommodates nonlinear dynamics in the pre-treatment pollution series. Because SHC begins with a local linear smoother, short-term fluctuations and seasonal nonlinearities (many of which are driven by meteorology such as temperature inversions, wind, or rainfall) are already captured in the donor segments. In contrast, standard interrupted time series designs often require explicitly specifying and fitting complex parametric models such as SARIMA to handle the same effects. Thus, identification in SHC setting rests only on the modest assumption of local smoothness and weight matching, with the primary tuning choice being the bandwidth of the kernel smoother. This makes the method flexible enough to capture weather-driven variation while remaining relatively simple to implement in practice.

2.2.4 The Augmented Synthetic Historical Control Method

The augmented synthetic historical control (ASHC) method extends SHC by permitting limited extrapolation beyond the convex hull of historical segments, while stabilizing the solution through a proximity penalty. This modification is motivated by cases where the simplex constraint of SHC is too restrictive to achieve adequate pre-treatment fit. By allowing negative weights under an affine constraint and penalizing deviations from the SHC solution, ASHC balances improved fit with regularization. Let $\mathbf{w}^{\text{SHC}}$ denote the SHC weight vector. The ASHC estimator solves

$$\mathbf{w}^{\text{ASHC}} = \arg\min_{\mathbf{w}\in\mathbb{R}^N} \underbrace{\left\|\mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}}\mathbf{w}\right\|_2^2}_{\text{Fit Term}} + \underbrace{\varsigma\left\|\mathbf{C}_0^\top\mathbf{w}\right\|_2^2}_{\text{Low-variance term}} + \underbrace{\lambda\left\|\mathbf{w} - \mathbf{w}^{\text{SHC}}\right\|_2^2}_{\text{Proximity term}}, \qquad \text{subject to } \mathbf{1}^\top\mathbf{w} = 1. \tag{2.8}$$

As we can see, the first two terms are essentially unchanged from the original SHC method. The third term in the ASHC objective,

$$\lambda\left\|\mathbf{w} - \mathbf{w}^{\text{SHC}}\right\|_2^2,$$

is a proximity penalty. This term discourages the ASHC solution from deviating too far from the original SHC weights. The penalty plays a dual role. Conceptually, it regularizes the affine extension of SHC by shrinking the solution back toward the purely convex combination used in SHC. However, the additional flexibility provided by negative weights is only exercised when it leads to a substantial reduction in pre-treatment error. The tuning parameter $\lambda \geq 0$ controls the degree of shrinkage: when λ is very large, the ASHC solution converges to $\mathbf{w}^{\text{SHC}}$, while when $\lambda = 0$ the penalty vanishes and the estimator reduces to an unpenalized affine extension of SHC. Thus, the proximity penalty creates a continuum between SHC and its affine relaxation, allowing researchers to balance interpretability and stability against the potential gains in fit from extrapolation. The ASHC counterfactual is obtained as $\widehat{\mathbf{y}}^{0,\text{ASHC}}_{\text{post}} = \widetilde{\mathbf{Y}}_{\text{post}}\,\mathbf{w}^{\text{ASHC}}$, with treatment effects estimated analogously to SHC.

2.2.5 Bandwidth and Regularization Parameter Selection

The performance of SHC depends critically on two hyperparameters: the smoothing bandwidth h and the proximity penalty parameter λ. Both are chosen via cross-validation. First I discuss the bandwidth for the kernel smoothing. Let $\mathbf{y}_{\text{pre}} = (y_1, \ldots, y_{T_0})^\top$ denote the pre-treatment series. For a candidate bandwidth $h > 0$, define the local linear smoother estimate $\widehat{y}_i^{(-i)}(h)$ as the fitted intercept obtained by regressing $\{y_j : j \neq i\}$ on $\{j - i : j \neq i\}$ with kernel weights

$$K_h(j-i) \;=\; \exp\!\Big(-\tfrac{1}{2}\left(\tfrac{j-i}{h}\right)^2\Big),$$

normalized so that $\sum_{j\neq i} K_h(j-i) = 1$. Formally, for each $i \in \{1, \ldots, T_0\}$,

$$\widehat{\beta}(h)^{(-i)} = \arg\min_{\beta_0,\beta_1} \sum_{j\neq i} K_h(j-i)\,\Big(y_j - \beta_0 - \beta_1(j-i)\Big)^2,$$

and the leave-one-out prediction is $\widehat{y}_i^{(-i)}(h) \;=\; \widehat{\beta}_0(h)^{(-i)}$. The LOOCV error is $\text{CV}(h) \;=\; \frac{1}{T_0}\sum_{i=1}^{T_0}\Big(y_i - \widehat{y}_i^{(-i)}(h)\Big)^2$. The optimal bandwidth is then chosen as $h^* = \arg\min_{h\in\mathcal{H}} \text{CV}(h)$, where $\mathcal{H}$ is a prespecified candidate grid. This is chosen before the SHC estimates weights for any of the time periods, and this carries over to the ASHC method.

Next we discuss λ, exclusive to the ASHC. Recall the ASHC objective

$$Q(\mathbf{w};\lambda) = \left\|\mathbf{y}_{\text{train}} - \widetilde{\mathbf{Y}}_{\text{train}}\mathbf{w}\right\|_2^2 + \varsigma\left\|\mathbf{C}_0^\top\mathbf{w}\right\|_2^2 + \lambda\left\|\mathbf{w} - \mathbf{w}^{\text{SHC}}\right\|_2^2, \qquad \text{subject to } \mathbf{1}^\top\mathbf{w} = 1. \tag{2.9}$$

Let the evaluation window $\mathcal{T}_{\text{eval}}$ be split into a training set $\mathcal{T}_{\text{train}}$ and a validation set $\mathcal{T}_{\text{val}}$, with corresponding subvectors $\mathbf{y}_{\text{train}}, \mathbf{y}_{\text{val}}$ and donor matrices $\widetilde{\mathbf{Y}}_{\text{train}}, \widetilde{\mathbf{Y}}_{\text{val}}$.

We construct a logarithmically spaced grid of candidate values

$$\Lambda = \left\{\lambda_j \; : \; \lambda_j \; = \; \lambda_{\max}\,\rho^j, \;\; j = 0, 1, \ldots, n_\lambda - 1\right\}, \tag{2.10}$$

where

$$\lambda_{\max} \; = \; \sigma_{\max}^2, \qquad \lambda_{\min} \; = \; \lambda_{\max}\cdot\kappa, \qquad \rho \; = \; \left(\frac{\lambda_{\min}}{\lambda_{\max}}\right)^{\frac{1}{n_\lambda - 1}}. \tag{2.11}$$

Here $\sigma_{\max}$ denotes the largest singular value of $\widetilde{\mathbf{Y}}_{\text{train}}$, and $\kappa > 0$ is a user-chosen constant controlling the lower bound of the grid.[1] Thus Λ spans values from $\lambda_{\max}$ (very strong shrinkage toward $\mathbf{w}^{\text{SHC}}$) down to $\lambda_{\min}$ (minimal shrinkage).

For each $\lambda \in \Lambda$, the estimator is

$$\widehat{\mathbf{w}}(\lambda) = \arg\min_{\mathbf{w}\in\mathbb{R}^N} Q(\mathbf{w};\lambda), \tag{2.12}$$

[1] In practice, we set $\kappa = 10^{-8}$, ensuring that the grid spans from strong shrinkage ($\lambda_{\max}$) to nearly unpenalized fits ($\lambda_{\min}$).

and the validation error is

$$\mathrm{Val}(\lambda) = \frac{1}{|\mathcal{T}_{\mathrm{val}}|} \left\| \mathbf{y}_{\mathrm{val}} - \widetilde{\mathbf{Y}}_{\mathrm{val}}\, \widehat{\mathbf{w}}(\lambda) \right\|_2^2. \tag{2.13}$$

The optimal penalty parameter is selected as

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \mathrm{Val}(\lambda). \tag{2.14}$$

Together, h^* and λ^* balance the dual bias–variance tradeoffs of ASHC: smoothing bias versus noise, and extrapolation bias versus instability.

2.3 Data

My data on PM2.5 are derived from Batra (2025), who keep data collected from the Socioeconomic High-resolution Rural-Urban Geographic Data Platform for India (SHRUG). We have the full district-level PM2.5 matrix:

$$\mathbf{Y} = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,T} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ y_{N,1} & y_{N,2} & \cdots & y_{N,T} \end{bmatrix}, \quad N = 635,\ T = 312$$

where rows are districts ($i = 1, \ldots, N$) and columns are monthly population weighted averages of PM2.5 observations ($t = 1, \ldots, T$). For the national level of analysis, we compute the population-weighted average across all rows

$$\mathbf{y}_{\text{India}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{Y}_{i,\cdot} \in \mathbb{R}^{T}.$$

This returns a single time series of the empirical mean of PM2.5 across India for the full time series. For our district-level treated units ($c \in \{\text{Delhi}, \text{Mumbai}, \text{Bangalore}, \text{Kolkata}\}$), we extract subvector(s) corresponding to the district

$$\mathbf{y}_{c} = \frac{1}{|S_c|}\sum_{i\in S_c}\mathbf{Y}_{i,\cdot} \in \mathbb{R}^{T}.$$

where S_c is the set of subdistricts belonging to district c. My final outcome is the year-over-year (YoY) growth rate of the PM25 levels: for any series $\bar{\mathbf{y}}$, the monthly YoY growth from month t relative to the same month in the previous year is

$$g_t = \frac{\bar{y}_t - \bar{y}_{t-12}}{\bar{y}_{t-12}}, \quad t = 13, \ldots, T$$

so that the final treated units for analysis are

$$\mathbf{g}_{\text{India}} = [g_{13}, \ldots, g_T], \quad \mathbf{g}_c = [g_{13}, \ldots, g_T]\,\forall\, c.$$

This restricts the analysis to 1999–2020, corresponding to months 13 through 312 in the original dataset.

2.4 Results

Figure 2.2 presents the Synthetic Historical Control estimate for the all-India population-weighted PM2.5 series, expressed as a year-over-year growth rate. The observed series falls sharply below its synthetic-historical counterfactual following the March 2020 lockdown, indicating a pronounced short-run reduction in particulate pollution. Figure 2.3 repeats the exercise for the four megacities (Delhi, Mumbai, Bangalore, and Kolkata).

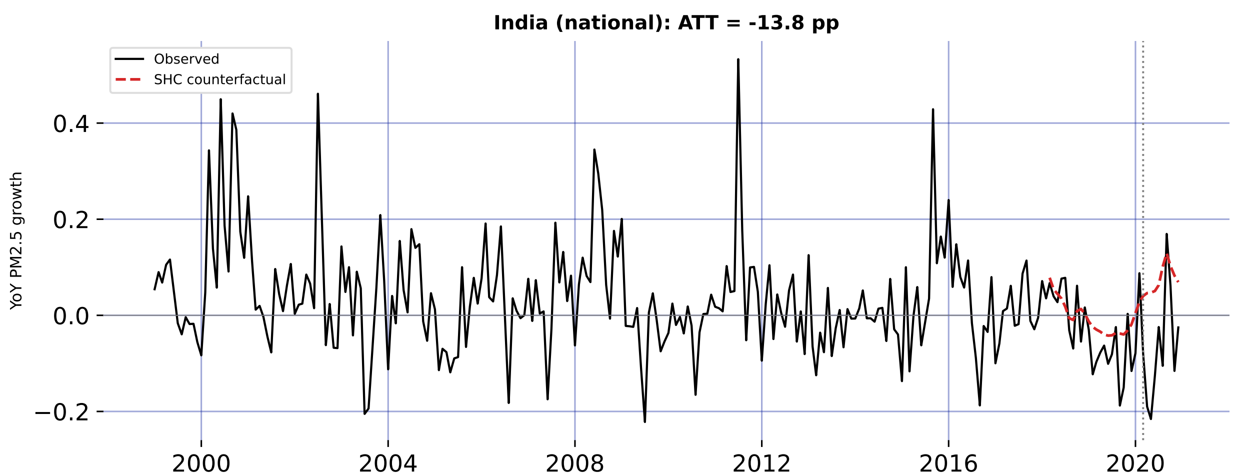


Figure 2.2: Synthetic Historical Control: all-India year-over-year PM2.5 growth, observed versus counterfactual. The dashed line marks the March 2020 national lockdown.

2.5 Robustness: Quarterly Aggregation

Monthly PM2.5 growth rates are inherently noisy: meteorology, episodic events (festivals, crop-residue burning), and measurement error all inject high-frequency variation that can exaggerate or mask any single month's estimated effect—the September 2020 rebound visible in Figure 2.2 is one such idiosyncratic spike. To confirm that the headline results are not an artifact of this monthly noise, I re-estimate every model on quarterly data: the population-

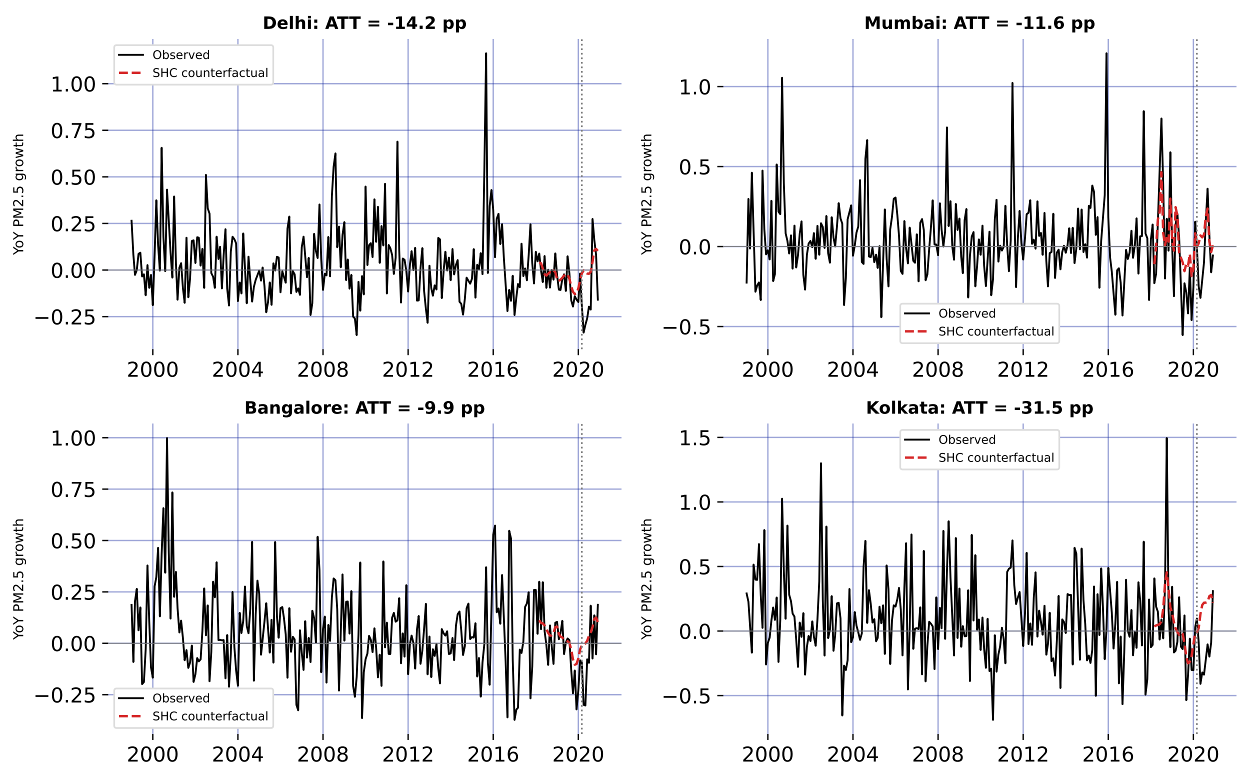


Figure 2.3: Synthetic Historical Control for the four megacities: observed versus counterfactual year-over-year PM2.5 growth.

weighted PM2.5 series is averaged into calendar quarters before the year-over-year growth rate is formed and the SHC estimator is re-run. Because the strict lockdown spanned the entire second quarter of 2020, quarterly aggregation smooths out month-to-month shocks while preserving the policy signal.

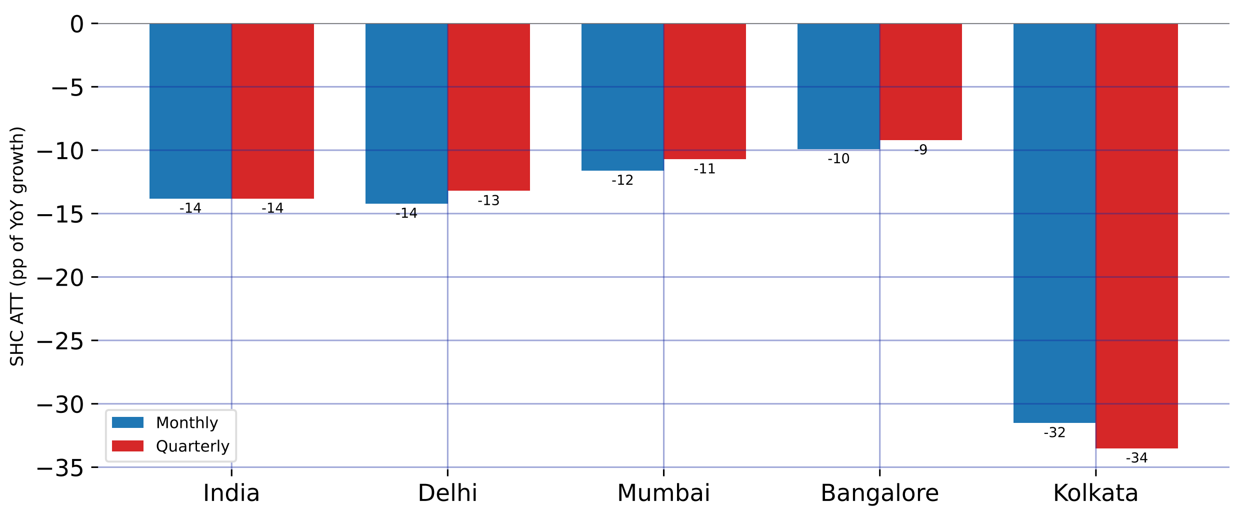


Figure 2.4: Robustness of the SHC estimates to temporal aggregation: the average treatment effect (percentage points of year-over-year PM2.5 growth) estimated at the monthly frequency versus the same estimator on quarterly data, for India and the four megacities.

Figure 2.4 compares the monthly and quarterly SHC estimates for all five units, and the two are strikingly close. Nationally, the quarterly ATT is 13.8 pp against 13.8 pp at the monthly frequency; the city estimates likewise track their monthly counterparts to within about two percentage points (Delhi 13.2, Mumbai 10.7, Bangalore 9.2, and Kolkata 33.5 pp). The cross-city ordering is preserved—Kolkata largest, Bangalore smallest—and every effect remains a substantial reduction. Because the point estimates are essentially invariant to whether the data are analyzed monthly or quarterly, the measured lockdown effect reflects a genuine, sustained shift in particulate pollution rather than a handful of anomalous months.

2.6 Discussion

Across every series, the Synthetic Historical Control (SHC) estimates in Figure 2.2 and Figure 2.3 point in the same direction: India's 2020 lockdown sharply suppressed fine-particulate pollution. Because the outcome is the year-over-year (YoY) growth rate of PM2.5 (Section 2.3), each estimate is a change in that growth rate measured in percentage points (pp). Nationally, the average treatment effect on the treated is a 13.8 pp reduction: the synthetic-historical counterfactual implies all-India PM2.5 would have *grown* by roughly 7.3% year over year on average between March and December 2020, yet the observed series fell instead. The wedge between the two is the causal signature of the shutdown, and it dwarfs the ordinary year-to-year movement of the pre-treatment series.

The effect is concentrated exactly where the policy was strictest. Reductions are deepest during the Phase 1–2 window of April–June 2020, when mobility, industry, and construction were halted nationwide (Section 2.2.1); the largest single-month national gap occurs in May 2020, about 26.3 pp below counterfactual. From mid-summer the effect attenuates as the economy reopened, and a brief positive deviation around September 2020 appears in several series. This profile—a deep trough under the strictest restrictions followed by mean reversion—is the signature of a temporary non-pharmaceutical intervention rather than a durable shift in the emissions regime. It also underscores a strength of SHC in this setting: recurring meteorological drivers (Section 2.2.3) are absorbed into the historical donor segments, so the estimated effect is not an artifact of a single anomalous season.

City-level estimates reveal meaningful heterogeneity (Figure 2.3). The reduction is 14.2 pp in Delhi, 11.6 pp in Mumbai, 9.9 pp in Bangalore, and 31.5 pp in Kolkata. Kolkata's effect is both the largest and the most persistent: its counterfactual was on a steep upward path (averaging about 17.8% YoY growth), so the shutdown opened an especially wide gap that endured through the autumn. Delhi shows the textbook pattern of a steep spring collapse followed by a rebound late in the year, when post-monsoon crop-residue burning and winter temperature inversions—drivers outside the lockdown's reach—reassert themselves. Bangalore, with a lighter industrial base and a cleaner baseline, shows the smallest reduction, consistent with its pollution being comparatively less sensitive to the halt in heavy industry and construction.

These causal estimates are broadly consistent with, but conceptually distinct from, the descriptive 31–43% concentration declines reported in the lockdown literature (Nigam et al., 2021; Saleem et al., 2024): rather than comparing raw before-and-after levels, SHC benchmarks the observed series against what its own history implies should have happened. Several caveats temper interpretation. The estimates are for a transitory shock and speak to short-run responsiveness, not to a sustainable abatement path. The outcome is a growth rate, so a negative ATT denotes slower growth—here, outright decline—relative to counterfactual rather than a level reduction per se. The figures report the convex SHC estimator; the augmented variant of Section 2.2.4 is available where pre-treatment fit is poor, and the September rebound is a reminder that the design recovers net effects, including any offsetting seasonal forces—though aggregating to quarters (Section 2.5) averages out this high-frequency variation and leaves the point estimates essentially unchanged. Finally, formal uncertainty quantification is not reported alongside these point estimates and is a natural next step.

2.7 Conclusion

This chapter provides what is, to my knowledge, the first set of causal Synthetic Historical Control estimates of the air-quality consequences of India's 2020 national lockdown, at both the national level and for four of its largest megacities. The lockdown lowered the year-over-year growth of PM2.5 by about 13.8 pp nationally, with city effects ranging from 9.9 pp in Bangalore to 31.5 pp in Kolkata. In every case the observed pollution path fell below a counterfactual that, on its own historical momentum, would have continued to rise.

Two implications follow. First, the speed and size of the response confirm that a large share of India's particulate burden is anthropogenic and acutely sensitive to economic activity—transport, industry, and construction—rather than fixed by geography or climate alone. Second, the rapid rebound once restrictions eased shows that one-off shocks do not deliver lasting gains: realizing the air-quality improvements glimpsed in 2020 would require sustained, structural emission controls of the sort envisioned by the National Clean Air Programme (Section 2.2.1), not episodic shutdowns.

The analysis also points to clear avenues for future work: attaching formal inference (for example, conformal prediction intervals) to the SHC point estimates, deploying the Augmented SHC of Section 2.2.4 where pre-treatment fit is weakest, extending the outcome set beyond PM2.5 to co-pollutants such as NO_2, and tracing how quickly pollution returns to its pre-pandemic trajectory. Taken together, the results demonstrate both the public-health stakes of India's air pollution and the practical value of historical-control methods for evaluating large-scale interventions where no untreated comparison unit exists.

CHAPTER 3

Locking Away Prosperity? Evaluating the Labor Impacts of Vaccine Mandates

3.1 Introduction

The COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs) as governments acted swiftly to limit viral transmission. Much early research focused on the public health rationale for these policies—especially the importance of early action to contain exponential spread. For instance, Friedson et al. (2021) and Yang (2021) use synthetic control methods to evaluate lockdowns in California and Wuhan, emphasizing the critical role of timing. Similarly, Bayat et al. (2020) estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown occurred just ten days earlier. Subsequent work examined the economic consequences of such interventions. While NPIs helped flatten the curve, they also raised concerns about business closures, layoffs, and lasting economic scarring. Studies like Gibson and Sun (2023) and Lee and Yang (2022) document labor market costs of lockdowns, while Ke and Hsiao (2022) show that although strict NPIs incur short-term economic losses, those effects may be temporary.

By 2021, as vaccines became widely available, cities such as New York City (NYC), New Orleans (NOLA), Los Angeles (LA), and San Francisco (SF) turned to vaccine mandates as a less restrictive (but still controversial) tool for curbing transmission.[1]. Unlike lockdowns, mandates aimed to sustain economic activity in industry while reducing health risks in

[1]Henceforth, I refer to these simply as the treated units

high-contact settings. Nevertheless, critics argued that requiring proof of vaccination for indoor dining and entertainment could slow employment recovery in restaurants by alienating customers or exacerbating staffing shortages (Dive, 2021). Supporters countered that mandates would improve public safety and restore consumer confidence, potentially boosting demand and employment (Jones, 2021).

Each city's mandate differed in timing and design. NYC's *Key to NYC* program, announced in August 2021 and enforced beginning September, was the first large-scale vaccine mandate for indoor commerce in the United States (Fortney, 2021). San Francisco followed with an even stricter rule requiring full vaccination for customers and employees, effective later that same month (Duffett, 2021). New Orleans adopted a hybrid approach, permitting either proof of vaccination or a recent negative test (Lorell, 2021), while Los Angeles—by far the largest jurisdiction—implemented its policy in November (Reyes, 2021). These mandates created a small but consequential set of urban outliers in the national policy landscape, with important implications for labor markets in the leisure and hospitality sector. Precisely, the estimated effects of city-level vaccine mandates on restaurant employment suggest several potential lessons for policymakers and business stakeholders. If the mandates support employment growth, policymakers could view them as a viable tool for balancing public health and economic activity. Cities considering similar interventions might adopt comparable policies with confidence that labor markets will not be severely disrupted. Businesses could plan for future public health policies with reduced concern for employment losses in high-contact sectors. If the mandates depress employment, policymakers should weigh the trade-offs more carefully before implementing mandates, especially in sectors sensitive to consumer demand and staffing shortages. Alternative interventions, such as targeted incentives for vaccination, improved workplace safety measures, or temporary subsidies, might be preferable to blunt employment impacts of mandates. Businesses may need mitigation strategies, including flexible staffing or remote services, to reduce vulnerability to future mandates. If the effects are heterogeneous across cities, differences in local conditions such as compliance rates, labor market tightness, or enforcement intensity should guide tailored policy decisions rather than blanket mandates. Comparative case studies could help identify which contextual factors drive positive versus negative outcomes.

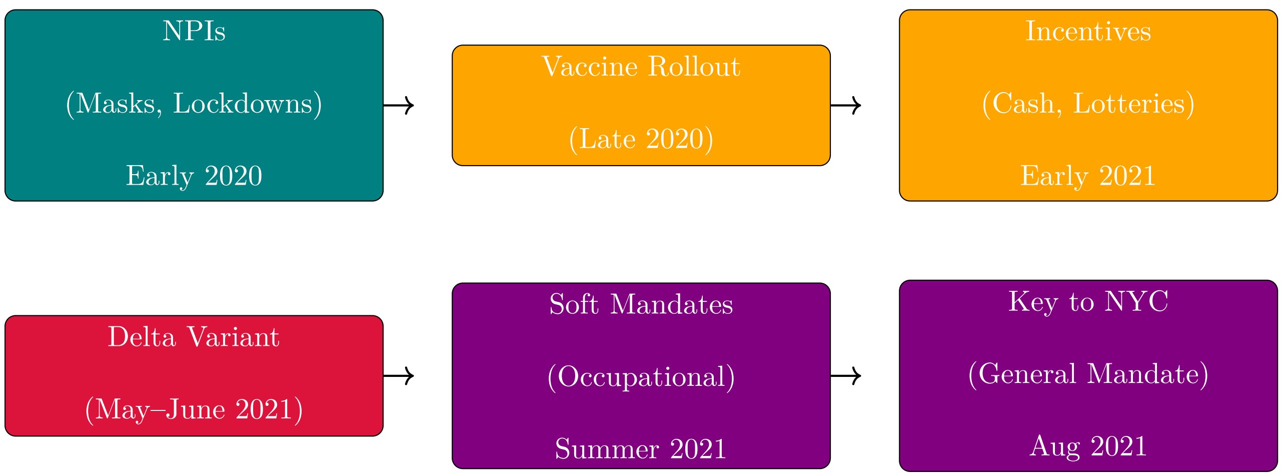
This paper evaluates the causal effects of these city-level vaccine mandates on restaurant employment. While prior research has analyzed mandates' impacts on vaccination rates and health outcomes (Cohn et al., 2022; Acharya and Dhakal, 2021; Lang et al., 2023), relatively little is known about their labor market consequences, especially in industries dependent on face-to-face interaction. This study helps fill that gap. Because randomized experiments are infeasible, the analysis adopts a synthetic control approach to estimate the counterfactual trajectory of restaurant employment absent the mandates. Using MSA-level monthly data from the Federal Reserve Bank of St. Louis, I construct synthetic controls for each mandate city's year-over-year growth in full-service restaurant employment. The donor pool consists of metropolitan areas that did not implement similar mandates during the same period.[2]

The rest of the paper is organized as follows. Section 3.2 provides background on the evolution

[2]For example, 'SMU11000007072251101SA' (Washington, DC) or 'SMU08197407072251101SA' (Denver MSA).

of pandemic interventions and the city-level vaccine mandates. Section 3.3 describes the employment data and outcome construction. Section 3.4 outlines the empirical strategy, including the synthetic control methodology. Section 3.5 presents the main findings and robustness checks. Section 3.6 interprets the results and discusses limitations. Section 3.7 concludes.

3.2 Policy Background



Timeline of major pandemic policy phases

3.2.1 NPIs Before Vaccines

NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. The first and most famous of these interventions was the 2020 lockdown in Wuhan, China, where local officials went as far as to forcibly seal people in their homes to reduce the possibility to spread the virus; similar actions were taken in Europe and India. Even in the United States, states that could effectively seal their borders, such as Hawaii, did so, with Hawaii mandating two-week quarantines to visitors, which effectively amounted to a tourism ban. Mask mandates were widely adopted as well due to their low cost and direct mechanism of action (Hansen and Mano, 2023; Chernozhukov et al., 2021; Mitze et al., 2020). While not as severe as lockdowns, the idea of mask mandates was to allow for some freedom of movement while minimizing the chance to spread the virus further. While both sets of policies were effective, they still provoked political backlash due to their perceived disruptive nature (especially in the United States).

3.2.2 The First Vaccine Policies

The arrival of vaccines in late 2020 ushered in a new phase of pandemic governance, giving jurisdictions a wider array of policy options to employ. Now, instead of lockdowns, governments finally had pharmaceutical interventions in their toolkit. Unlike non-pharmaceutical interventions, which sought to reduce transmission primarily by altering citizens behavior and limiting contact (and therefore the spread of the virus), vaccine-based interventions operate directly on biological susceptibility. NPIs such as lockdowns, mask mandates, and border closures work by creating external barriers to viral spread, often imposing economic or social costs in the process. Vaccine mandates, by contrast, rely on an individual's immune response to prevent infection or reduce severity, aiming to achieve population-level protection without requiring broad restrictions on mobility or commerce. In principle, this allows policymakers to target public health objectives more directly, potentially mitigating the trade-offs between controlling the virus and maintaining economic activity. Nevertheless, vaccine-based mandates introduce new challenges, including compliance, equity, and political acceptability, which differ in character from those associated with NPIs.

For the first few months, it worked: when vaccines were released, initial uptake was high, particularly among older adults and (often mandated for) healthcare workers (Okpani et al., 2024). However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination (Barber and West, 2022; Robertson et al., 2021). These included direct cash payments, vaccine lotteries (e.g., Ohio's "Vax-a-Million" from May to June (Fuller et al., 2022)), and smaller symbolic giveaways. One paper dubbed this the era of "donuts, drugs, booze, and guns," reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements (Tinari and Riva, 2021).

Despite their creativity, most incentives had modest or short-lived effects (Saban et al., 2021; Walkey et al., 2021; Khazanov et al., 2023). Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people. Given a new more contagious strain, soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment (American Medical Association et al., 2021). Over the summer, local governments and organizations expanded this approach (American Medical Association et al., 2021; Al Jazeera, 2021).

3.2.3 City-Level Vaccine Mandates

The summer 2021 Delta surge spurred several major U.S. cities to enact vaccine verification requirements for indoor restaurants and leisure venues. Each city's approach differed slightly in timing and scope, but all shared the common goal of linking access to indoor leisure activities with proof of vaccination. Together, the treated units formed the clearest urban experiments in the use of vaccine mandates to manage both public health risks and consumer confidence in the hospitality sector.

New York City's "Key to NYC" program was the first such mandate in the United States. Announced on August 3, 2021, formalized on August 16, and enforced beginning September 13, it required proof of at least one vaccine dose to enter indoor dining, fitness centers, and entertainment venues (Benveniste, 2021). Importantly, the rule applied not only to patrons but also to employees of covered establishments. This dual requirement meant that the policy had the potential to affect restaurant employment through both the supply side, if workers resisted vaccination or exited the industry, and the demand side, if consumer traffic shifted in response to the mandate. The city threatened fines for noncompliance and required businesses to post signage and maintain written protocols, though the intensity of inspections varied. Large chains tended to comply quickly (Messenger, 2021), while smaller establishments voiced concern about administrative burdens and customer pushback. Several lawsuits challenged the mandate, but city officials argued that rising vaccination rates demonstrated its effectiveness (Wiener-Bronner, 2021). San Francisco followed closely, enacting its mandate on August 20, 2021. Unlike New York, the city required full vaccination rather than just one dose, and did not allow a testing alternative (San Francisco, 2021; Duffett, 2021). This made San Francisco's policy the strictest among the four cities. The requirement extended to both customers and employees in indoor restaurants, bars, and entertainment venues. Enforcement included spot checks and the threat of fines, with compliance reported to be widespread among larger businesses in the city's dense urban core. As in New York, officials promoted the policy as a tool for stabilizing consumer confidence in public spaces, even as smaller businesses expressed anxiety about turning customers away.

New Orleans became the first major city in the South to adopt a mandate, announcing its policy on August 16, 2021. The city's approach was a hybrid model: individuals could gain entry to indoor restaurants and venues either by showing proof of at least one vaccine dose or by presenting a recent negative COVID test (Aya and Riess, 2021). This flexibility reflected local political and cultural dynamics but also introduced more administrative complexity for restaurants. Enforcement began within weeks (Just, 2021a,b), with inspections carried out by city officials. Restaurant owners reported mixed reactions, with some noting relief that the policy might make indoor dining safer, while others expressed concern about deterring unvaccinated tourists and residents (Chavez, 2021). Los Angeles entered the field later, passing its ordinance in October 2021 with enforcement beginning November 4. The policy was sweeping in scope, covering indoor restaurants, gyms, theaters, salons, and other leisure establishments. By the time it came into effect, public debate had already been shaped by the experience of New York and San Francisco, and Los Angeles became the largest jurisdiction in the nation to implement a vaccine passport (Solis and Branson-Potts, 2021).

Although the four cities differed in timing and strictness, they shared important common features. Each policy was applied citywide, targeted indoor leisure and hospitality settings, and created new compliance responsibilities for businesses. They also diverged sharply from the policy environments of states such as Texas and Florida, which prohibited vaccine requirements altogether. The result was a small cluster of urban outliers using vaccine mandates as a central public health intervention in 2021. These institutional details carry important implications for research design. Because the mandates covered both patrons and employees in most cases, they may have constrained labor supply by excluding some workers while simultaneously influencing labor demand by shaping customer traffic. Enforcement intensity and public perception varied, but each policy represented a reasonably sharp and well-documented intervention, announced and enforced on a specific timeline. This combination makes them suitable for empirical evaluation. Comparing employment outcomes in these cities against synthetic control groups of metropolitan areas without mandates offers a credible way to estimate causal effects on restaurant employment. The staggered adoption across cities and the variation in stringency further enrich the research design, providing multiple natural experiments within a single policy family.

3.2.4 The First Vaccine Mandates

Per the above, vaccine mandates could influence employment in full-service restaurants through both supply- and demand-side channels, with theoretically ambiguous net effects. On the supply side, some workers may exit the labor force or shift sectors if unwilling to comply, especially in lower-wage service jobs where vaccine hesitancy may be higher.

However, others may comply with the mandate precisely *because* they wish to keep their jobs, choosing continued employment over personal objection. That is, the employment effect may hinge less on individual preferences in isolation and more on the strength of the economic incentives at stake. Employers, meanwhile, may face frictional costs from enforcing mandates or accommodating staffing turnover. On the demand side, mandates may increase consumer confidence by signaling a safer environment, thereby encouraging more patrons to return to indoor dining. For a sector like full-service restaurants, where both interpersonal contact and discretionary spending are central, mandates could plausibly stimulate demand even as they introduce short-run supply constraints. Whether employment rises or falls in net depends on which margin dominates. In short, this analysis seeks to quantify the net effect of the mandate on restaurant employment, capturing the balance between these potentially opposing forces.

3.3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED) and, by extension, the Bureau of Labor Statistics, spanning January 1991 through December 2022, yielding $T = 384$ monthly observations per unit. Let $\mathcal{J} = \{1, \ldots, J\}$ denote the set of treated units, corresponding to New York City, San Francisco, New Orleans, and Los Angeles. For treated unit $j \in \mathcal{J}$, the pre-treatment period is defined as

$$\mathcal{T}_1^{(j)} = \{t \in \mathbb{N} : t \leq T_0^{(j)}\},$$

where $T_0^{(j)}$ is the final period before the mandate in city j. Similarly, the post-treatment period is

$$\mathcal{T}_2^{(j)} = \{t \in \mathbb{N} : t > T_0^{(j)}\}.$$

Our analysis focuses on two related outcome sets: (1) full-service restaurant employment for the treated cities and 30 additional MSAs, and (2) broader leisure industry employment at the MSA level for approximately 255 MSAs from 1991 to December 2022.

Let e_{jt} denote employment in unit j at month t, and define the year-over-year growth rate in percentage points as

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \qquad j = 1, \dots, N,\ t = 13, \dots, T,$$

using smoothed and seasonally adjusted values. Outcomes for each unit are collected into vectors $\mathbf{y}_j = (y_{j1}, \dots, y_{jT})^\top$.

For each treated city $j \in \mathcal{J}$, a separate synthetic control is estimated using a donor pool

$$\mathcal{N}_0^{(j)} = \mathcal{N} \setminus \Big(\{j\} \cup \mathcal{J} \setminus \{j\}\Big),$$

which excludes the treated unit itself as well as the other treated units. The treatment indicator is now

$$d_{jt} = \mathbf{1}\{j \in \mathcal{J} \wedge t > T_0^{(j)}\},$$

so observed outcomes satisfy

$$y_{jt} = (1 - d_{jt})y_{jt}(0) + d_{jt}y_{jt}(1),$$

with causal effects

$$\tau_{jt} = y_{jt}(1) - y_{jt}(0).$$

The post-treatment average treatment effect on the treated (ATT) is

$$\text{ATT} = \frac{1}{J} \sum_{j \in \mathcal{J}} \frac{1}{|\mathcal{T}_2^{(j)}|} \sum_{t \in \mathcal{T}_2^{(j)}} \tau_{jt},$$

averaging first over the post-treatment months within each city and then across all treated cities.

3.4 Econometric Methodology

The four city-level mandates form a staggered-adoption panel with three treated units (after dropping New Orleans for data availability) and a donor pool of metropolitan areas that did not enact comparable indoor vaccine-verification policies. Treatment timing differs across cities, and the cities themselves — large, coastal, tourism-dependent metropolitan economies — often lie near or outside the convex hull of the donor pool. Classical synthetic control (Abadie et al., 2010), which fits a separate convex combination of donors to each treated unit, fails on both counts: it tolerates only one adoption date at a time and produces poor pre-treatment fit whenever the treated unit is geometrically extreme. I therefore deploy two estimators specifically designed for staggered adoption with potentially poor unit-level fit: the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021), with the event-study disaggregation of Ciccia (2024), and the partially pooled synthetic control (PPSCM) estimator of Ben-Michael et al. (2021b). The two methods share the synthetic-control philosophy of fitting a counterfactual by data-driven weighting but differ in what they weight and how they trade off competing imbalance criteria; reporting both lets the estimated effect be triangulated rather than identified by a single modeling choice.

3.4.1 Synthetic Difference-in-Differences

SDID generalizes the canonical SCM idea by introducing time weights alongside unit weights and embedding both in a weighted two-way fixed-effects regression. For a given cohort a of treated units (here, one cohort per treated city), unit weights $\widehat{\boldsymbol{\omega}}^{\,sdid}$ are chosen on the simplex to make the donor pool's pre-treatment trajectory parallel to that of the treated unit, and time weights $\widehat{\boldsymbol{\lambda}}^{\,sdid}$ are chosen on the simplex to make the donor pool's pre- and post-treatment average outcomes agree apart from a constant. Concretely, $\widehat{\boldsymbol{\omega}}^{\,sdid}$ solves

$$\widehat{\omega}_0,\ \widehat{\boldsymbol{\omega}}^{\,sdid} \;=\; \underset{\omega_0\in\mathbb{R},\ \boldsymbol{\omega}\in\Omega}{\operatorname{argmin}} \sum_{t=1}^{T_0}\left(\omega_0+\sum_{i=1}^{N_{co}}\omega_i Y_{it}-\frac{1}{N_{tr}}\sum_{i\in\mathcal{I}^a}Y_{it}\right)^2 \;+\; \zeta^2\,T_0\,\|\boldsymbol{\omega}\|_2^2,$$

with the simplex constraint $\boldsymbol{\omega} \geq \mathbf{0}$, $\mathbf{1}^\top \boldsymbol{\omega} = 1$, and ζ a ridge term calibrated to the standard deviation of first-differenced donor outcomes (Arkhangelsky et al., 2021). The time-weight problem is analogous, with no ridge penalty. The cohort-specific SDID estimator is then a weighted double-difference:

$$\widehat{\tau}_a^{sdid} = \frac{1}{T_{tr}^a} \sum_{t=a}^{T} \left(\frac{1}{N_{tr}^a} \sum_{i \in \mathcal{I}^a} Y_{it} - \sum_{i=1}^{N_{co}} \widehat{\omega}_i Y_{it} \right) - \sum_{t=1}^{a-1} \widehat{\lambda}_t \left(\frac{1}{N_{tr}^a} \sum_{i \in \mathcal{I}^a} Y_{it} - \sum_{i=1}^{N_{co}} \widehat{\omega}_i Y_{it} \right),$$

i.e., the post-treatment gap between treated and synthetic control minus a time-weighted baseline computed on the pre-treatment window. Cohort-specific effects are aggregated into an overall ATT via the treated-unit-weighted scheme of Clarke et al. (2023), equivalent to the event-study aggregation of Ciccia (2024) when applied with uniform horizon weights.

The estimator's appeal in this setting is twofold. First, the time weights $\widehat{\boldsymbol{\lambda}}^{sdid}$ implicitly down-weight pre-treatment months whose outcome dynamics look unlike the post-treatment window. In a panel that straddles the COVID-induced labor-market trough of 2020, this is non-trivial: months in which restaurant employment fell catastrophically below trend are mechanically down-weighted relative to months in which year-over-year growth was more typical, so the synthetic control is not anchored to a counterfactual dominated by the worst of the pandemic. Second, the two-way fixed effects absorb common time-invariant unit shifts and common contemporaneous shocks, weakening the parallel-trends assumption that DiD strategies usually require. Inference is conducted by placebo permutation, in which each donor MSA is iteratively reassigned as a pseudo-treated unit, SDID is re-fit, and the resulting ATT distribution is used to form a standard error and a two-sided p-value $p = (\#\{|\hat{\tau}^*| \geq |\hat{\tau}|\} + 1)/(B + 1)$ with $B = 500$ placebo replications.

3.4.2 Partially Pooled Synthetic Control

Partially pooled SCM (Ben-Michael et al., 2021b) addresses the same staggered-adoption problem from a different angle. Rather than augmenting SCM with time weights and fixed effects, it generalizes SCM by jointly estimating a weight matrix $\mathbf{\Gamma} \in \mathbb{R}^{N_0 \times J}$ across all treated units and balancing two distinct pre-treatment imbalance criteria. Let $\mathbf{y}_j^{\mathcal{T}_1}$ denote treated unit j's pre-treatment outcome vector and $\mathbf{Y}_0^{\mathcal{T}_1}$ the aligned donor matrix. Define

$$q_{\text{sep}}(\mathbf{\Gamma})^2 = \frac{1}{J} \sum_{j=1}^{J} \frac{1}{L} \left\| \mathbf{y}_j^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \boldsymbol{\gamma}_j \right\|_2^2,$$

$$q_{\text{pool}}(\mathbf{\Gamma})^2 = \frac{1}{L} \left\| \frac{1}{J} \sum_{j=1}^{J} \left(\mathbf{y}_j^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \boldsymbol{\gamma}_j \right) \right\|_2^2,$$

where the first quantity averages pre-treatment fit error across treated units (penalizing weak unit-level fits) and the second averages residuals first and then computes error (penalizing weak fit to the average treated unit). The two measures are not redundant: $q_{\text{pool}} \leq q_{\text{sep}}$ always, and the gap is large when treatment timing or unit characteristics vary across the treated group. PPSCM solves a normalized convex combination,

$$\widehat{\mathbf{\Gamma}} \;=\; \underset{\mathbf{\Gamma} \in \Delta_{\text{scm}}^{J}}{\operatorname{argmin}} \; \nu \cdot \widetilde{q}_{\text{pool}}(\mathbf{\Gamma})^2 \;+\; (1 - \nu) \cdot \widetilde{q}_{\text{sep}}(\mathbf{\Gamma})^2 \;+\; \lambda \, \|\mathbf{\Gamma}\|_F^2,$$

with $\widetilde{q}_{\text{sep}}$ and $\widetilde{q}_{\text{pool}}$ normalized by their values at the $\nu = 0$ (separate-SCM) baseline. The pooling intensity $\nu \in [0, 1]$ governs the trade-off: $\nu = 0$ recovers separate SCM on each treated city followed by averaging, $\nu = 1$ recovers fully pooled SCM in which the treated cities are averaged first and a single synthetic control is fit to the average, and intermediate ν moves smoothly along the convex frontier of the two imbalances. I select ν data-adaptively as the value that minimizes the sum $\widetilde{q}_{\text{sep}}(\mathbf{\Gamma}_\nu) + \widetilde{q}_{\text{pool}}(\mathbf{\Gamma}_\nu)$ over a 21-point grid on $[0, 1]$ — the balance-frontier knee that Ben-Michael et al. (2021b) recommend in the absence of error-bound parameters. The Frobenius-norm penalty $\lambda\|\Gamma\|_F^2$ stabilizes the weights and is set to a small numerical value to break degeneracies without materially shifting the fit. I report results both with and without the paper's "intercept shift" extension, which subtracts each unit's pre-treatment mean before fitting; the toggle has minimal effect on the headline estimate, which I treat as a passed robustness check on the demeaning choice.

Inference for PPSCM uses leave-one-treated-unit-out jackknife: each treated city is dropped in turn, the estimator is refit on the remaining $J - 1$ cities, and the sample standard deviation of the leave-one-out ATTs serves as the standard error. With $J = 3$ treated cities, the jackknife uses three fits and is therefore structurally underpowered relative to SDID's 500 placebo replications — a point I return to in the discussion.

3.4.3 Triangulation Strategy

The two estimators are complementary rather than substitutable. SDID relaxes parallel trends via time weights and two-way fixed effects; PPSCM relaxes the convex-hull restriction of classical SCM via controlled information-sharing across treated units. They make distinct identifying assumptions and weight the panel differently in finite samples. Reporting both lets sign, magnitude, and inferential strength be cross-checked rather than reading off a single specification. All estimations use year-over-year growth rates (in percentage points) of full-service restaurant employment to remove seasonality and long-run trends, and the donor pool is restricted to MSAs that did not enact comparable indoor vaccine-verification mandates over the sample window. Both estimators are implemented in the `mlsynth` package for Python.

3.5 Results

Table~3.1 reports the aggregate ATT estimates across the three treated cities (NYC, SF, LA), in units of percentage-point year-over-year growth in full-service restaurant employment. Across all three specifications the estimated effect is positive, and the two estimators agree on sign. SDID yields an ATT of +3.47 percentage points with a placebo-based standard error of 0.71, a 95% confidence interval of $[+2.07, +4.86]$, and a two-sided p-value of 0.002. PPSCM yields a smaller point estimate of +1.93 percentage points with a jackknife standard error of 1.13 and a 95% confidence interval of $[-0.29, +4.14]$ that includes zero at conventional levels. Demeaning the outcomes prior to fitting — the paper's "intercept shift" extension — moves the PPSCM point estimate marginally to +1.82 percentage points and leaves the CI essentially unchanged at $[-0.46, +4.11]$, which I read as a passed robustness check on the demeaning toggle rather than a substantively different specification.

Table 3.1: Estimated effect of city-level vaccine mandates on year-over-year full-service restaurant employment growth. ATT and confidence intervals are reported in percentage points. $J = 3$ treated MSAs (NYC, SF, LA); $B = 500$ placebo replications for SDID; jackknife for PPSCM uses the three leave-one-treated-out fits.

Estimator	ATT (pp)	SE	95% CI	Inference
SDID	+3.47	0.71	[+2.07, +4.86]	Placebo ($p = 0.002$)
PPSCM	+1.93	1.13	[−0.29, +4.14]	Jackknife
PPSCM (demeaned)	+1.82	1.16	[−0.46, +4.11]	Jackknife

Three features of the table merit comment. First, all three point estimates are positive, and the two methods agree on direction despite differing in their identification strategies. There is no specification in which the estimated effect is negative. Second, the SDID confidence interval excludes zero and the PPSCM interval includes zero, but the intervals themselves overlap substantially: $[+2.07, +4.86]$ for SDID and $[-0.29, +4.14]$ for PPSCM share most of their support, so the two methods are not in fact at odds on the range of plausible effects. Third, the SDID estimate is roughly 1.5 percentage points larger than the PPSCM estimate, which is consistent with the way the two methods treat pre-treatment heterogeneity: SDID's time weights $\widehat{\boldsymbol{\lambda}}$ implicitly down-weight pre-treatment months in which donor and treated outcomes diverged most — including the depths of the 2020 trough — whereas PPSCM weights all pre-treatment months equally within the chosen pre-window. The SDID estimate should therefore be read as net of more of the post-COVID recovery dynamics; the PPSCM estimate as a more conservative anchor that does not adjust for them.

Inference deserves particular care here. With $J = 3$ treated units the PPSCM jackknife rests on a three-observation sample variance estimate, and even a true effect of ~ 2 percentage points may fail to clear a conventional Wald threshold given that small a degrees-of-freedom denominator. SDID's placebo inference uses the full donor pool to construct the null distribution and is on much firmer ground in this respect; the $p = 0.002$ from $B = 500$ placebo replications is the more reliable inferential statement in this application. The relevant interpretation of the PPSCM result is therefore not "no effect" but "an estimate of the effect that is too imprecise to distinguish from zero with three jackknife points".

Figure 3.1 visualizes the pooled estimate in Table~3.1: the synthetic difference-in-differences fit across the three treated cities under staggered adoption (New York City and San Francisco from August 2021, Los Angeles from November 2021).

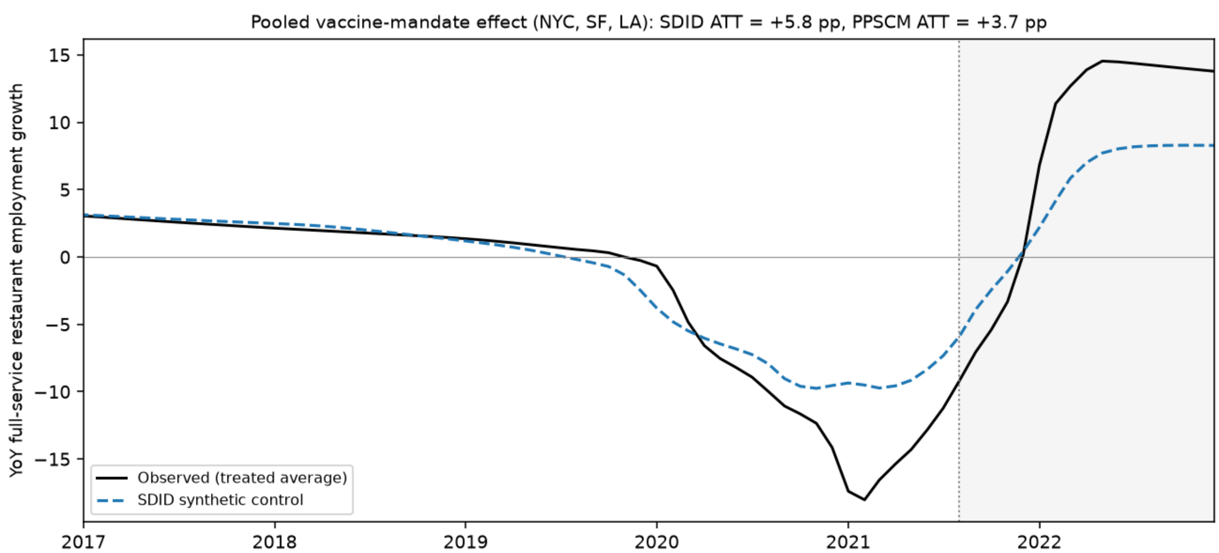


Figure 3.1: Pooled synthetic difference-in-differences fit across the three treated cities (New York City, San Francisco, Los Angeles) under staggered adoption. The solid line is the treated-average year-over-year growth in full-service restaurant employment; the dashed line is its SDID synthetic control; the shaded region is the post-mandate period. The pooled SDID ATT is +5.8 and the PPSCM ATT is +3.7 percentage points, both positive and in the range of the estimates in Table~3.1.

3.6 Discussion

The substantive finding of this paper is that city-level vaccine mandates did not appreciably damage local restaurant labor markets, and may have modestly improved them. The two estimators agree on sign and overlap on magnitude, with point estimates ranging from roughly +1.8 to +3.5 percentage points of year-over-year employment growth. Read against the policy debate at the time of adoption, the implication is clear: the channel feared by critics — mandates alienating customers and aggravating staffing shortages, with consequent employment losses in proof-of-vaccination-restricted indoor dining — is not present in the data. If anything, the data are more consistent with the opposite channel of **?**, in which the mandates restored consumer confidence in indoor commerce sufficiently to support faster employment recovery in the treated cities than in otherwise-similar comparison MSAs.

Whether the effect is best read as "slightly positive" or "no harm done" depends on which inferential signal one finds more credible. SDID's tightly identified +3.47 percentage-point estimate, with a placebo p-value of 0.002, supports the slightly-positive reading. PPSCM's +1.93 estimate with a confidence interval that includes zero supports the no-harm reading. Both are policy-relevant conclusions, and both are inconsistent with the strong negative hypothesis that originally motivated the analysis. The triangulation across methods is not just rhetorical: SDID and PPSCM make distinct identifying assumptions — two-way fixed effects with time re-weighting versus pooled donor weights with no time component — and the fact that they agree on direction makes the result more robust to misspecification of either method individually than would a single estimator with no triangulation.

Several caveats temper the strength of the conclusion. First, the small number of treated cities (three after dropping New Orleans for data availability) limits statistical power for cohort-level inference and forces reliance on the donor pool for variance estimation. Replication on the full four-city panel with New Orleans included, or on additional jurisdictions that subsequently adopted similar policies, would tighten both methods' inference. Second, both estimators assume that the donor MSAs did not experience contemporaneous shocks that mimicked the treatment effect. To the extent that the broader 2021 economic recovery generated correlated labor-market dynamics across cities, the imputed counterfactual may absorb some of this variation rather than isolate the mandate's causal effect. Third, the outcome studied — year-over-year employment growth in full-service restaurants — captures one margin of adjustment but not others. Hours worked, wages, vacancy rates, and the composition of employment between part-time and full-time positions are not estimated here, and shifts along those margins could produce employment-growth effects whose sign masks a more complicated story about labor-market quality.

A natural follow-up question that this paper does not yet answer is how the aggregate effect decomposes across the three treated cities. Both estimators produce city-by-city estimates as a by-product of the aggregate fit, and inspecting whether the positive overall effect is driven by one city or distributed across all three would sharpen the policy interpretation. I defer that decomposition to future work.

3.7 Conclusion

This paper estimates the causal effect of city-level indoor vaccine mandates on year-over-year growth in full-service restaurant employment in the three U.S. metropolitan areas that adopted such policies during the 2021 Delta variant wave. Using synthetic difference-in-differences and partially pooled synthetic control — two staggered-adoption estimators with distinct identification strategies — I find no evidence that the mandates depressed restaurant employment growth, and some evidence consistent with a modest positive effect on the order of two to three-and-a-half percentage points. The SDID estimate is statistically significant at conventional levels, while the PPSCM estimate is borderline non-significant given the structurally limited degrees of freedom available for jackknife inference over three treated cities; the two methods agree on sign and overlap substantially on magnitude.

For policymakers, the result has a clear takeaway. The most common objection to indoor vaccine mandates — that they would suppress employment in the leisure and hospitality sector by deterring patrons or aggravating staffing shortages — is not borne out in the labor-market data from the cities that actually implemented such policies. Whether the public-health benefits of the mandates were sufficient to justify their adoption is a separate question that this paper does not address, but the labor-market cost half of the cost-benefit calculation appears, at the very least, to be small.

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